Robotics Assignment

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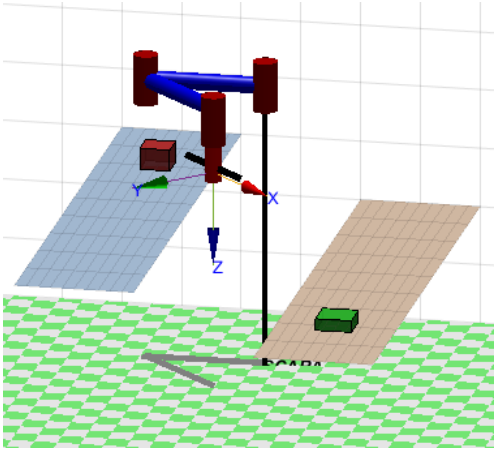
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Abstract

This report presents the design, kinematic analysis, and simulation of an automated industrial assembly line utilizing a 4-DOF SCARA robotic manipulator. The primary objective is the dynamic pick-and-place assembly of two parts (T_A and T_B) transported via independent conveyor belts moving at different velocities.In the first part of the project, the deterministic framework is established. Homogeneous transformation matrices are derived to define the precise operational sequences, including synchronization with the moving conveyor belt A for a secure grasping phase and the subsequent alignment and 90° screwing assembly onto the stationary part B. Continuous Cartesian trajectories with smooth velocity profiles are generated to ensure collision-free transitions between frames, complying with strict timing constraints. The system's behavior is simulated and validated using the Matlab Robotics Toolbox.In the second part, manufacturing uncertainties are introduced by injecting stochastic spatial (δ_x, δ_y) and rotational (θ_δ) deviations into the initial placement of part B. To accommodate these variations without re-engineering the trajectory profiles from scratch, Dynamic Movement Primitives (DMPs) with spatial and rotational scaling are implemented. The DMPs are trained using the nominal trajectories from Part 1 as baseline demonstrations.Multi-cycle simulations confirm that the DMP framework successfully adapts the end-effector paths in real-time, ensuring robust assembly execution across varying perturbation profiles. Detailed kinematic profiles—illustrating end-effector and joint positions and velocities—are presented alongside comparative analysis between the demonstrated and adapted trajectories, demonstrating the efficacy of learning-based control in adaptive industrial automation.

Part 1 - Kinematic Modelling and Cartesian Trajectory Design

This document presents the mathematical analysis for Requirements 1 and 2 of Part 1. All quantities are expressed in the inertial (world) frame W, located at $z = 0$, in SI units. The development follows exactly the model and the trajectory generator implemented.



Robot model and kinematics

Denavit–Hartenberg parameters

The manipulator is the 4-DoF SCARA returned by `create_robot`: two revolute joints (q_1, q_2) forming a planar arm, one prismatic joint (q_3) for the vertical motion, and a revolute wrist (q_4). The standard DH table is

Link i	θ_i	d_i	a_i	α_i
1 (R)	q_1	0	1	0
2 (R)	q_2	0	1	π
3 (P)	0	q_3	0	0
4 (R)	q_4	0.25	0	0

with base transform $\mathbf{T}_{\text{base}} = \text{Transl}(0, 0, 2.4)$ and tool transform $\mathbf{T}_{\text{tool}} = \text{Transl}(0, 0, \ell_g + \frac{1}{2}\ell_f) = \text{Transl}(0, 0, 0.2)$, where $\ell_g = 0.1$ (palm offset) and $\ell_f = 0.2$ (finger length). The tool transform places the TCP *between* the two fingers, at half the finger length from the palm, as required.

Forward kinematics

Because $d_1 = d_2 = 0$ and $a_1 = a_2 = 1$, the first two joints behave as a planar $2R$ arm in the horizontal plane carried at height 2.4. The wrist position is therefore

$$\begin{aligned} x &= \cos q_1 + \cos(q_1 + q_2), \\ y &= \sin q_1 + \sin(q_1 + q_2). \end{aligned} \tag{1}$$

The twist $\alpha_2 = \pi$ flips the z axis downward, so the prismatic joint and the wrist offsets lower the TCP. Collecting the constant vertical offsets,

$$z = 2.4 - q_3 - \underbrace{0.25}_{d_4} - \underbrace{0.20}_{\text{tool}} = 1.95 - q_3. \tag{2}$$

Orientation (single rotational DoF). The accumulated tool rotation is

$$\mathbf{R}_{\text{tool}} = \mathrm{R}_z(q_1)\,\mathrm{R}_z(q_2)\,\mathrm{R}_x(\pi)\,\mathrm{R}_z(q_4) = \mathrm{R}_z(q_1 + q_2 - q_4)\,\mathrm{R}_x(\pi), \tag{3}$$

where the identity $\mathrm{R}_x(\pi)\mathrm{R}_z(\psi) = \mathrm{R}_z(-\psi)\mathrm{R}_x(\pi)$ was used. The factor $\mathrm{R}_x(\pi)$ shows that the tool z-axis always points along $-z_W$ (the gripper approaches the parts from above), confirming *Hint 1*: the only orientation freedom is a rotation about the vertical axis. We parametrise it by the single angle

$$\boxed{\varphi \;=\; q_1 + q_2 - q_4} \tag{4}$$

so that $\mathbf{R}_{\text{tool}} = \mathrm{R}_z(\varphi)\mathrm{R}_x(\pi)$, i.e. a rotation of $-\varphi$ about $-z_W$ composed with the fixed downward flip. Hence every reachable pose is fully described by the four task coordinates (x, y, z, φ), and orientation trajectories reduce to scalar trajectories in φ.

Inverse kinematics (closed form)

Given a task pose (x, y, z, φ), inverting (1)–(4) gives

$$\begin{aligned} D &= \frac{x^2 + y^2 - 2}{2}, & q_2 &= \operatorname{atan2}\big(\sqrt{1 - D^2},\, D\big), \\ q_1 &= \operatorname{atan2}(y, x) - \operatorname{atan2}\big(\sin q_2,\, 1 + \cos q_2\big), & q_3 &= 1.95 - z, \\ q_4 &= q_1 + q_2 - \varphi. \end{aligned} \tag{5}$$

Comment. The positive square root in q_2 selects the *elbow-up* branch ($q_2 \in (0, \pi)$). This branch is chosen because it is the one consistent with the supplied home configuration $\boldsymbol{q}_0 = [-\frac{\pi}{2},\ \frac{3\pi}{4},\ 0.1,\ 0]^\mathsf{T}$ ($q_2 = 3\pi/4 > 0$); keeping a single branch throughout the cycle guarantees a continuous joint path with no elbow re-configuration (which would otherwise produce jumps in the joint plots).

Requirement 1 — Homogeneous transforms of the intermediate poses

Substituting $\mathbf{R}_{\text{tool}} = \mathrm{R}_z(\varphi)\mathrm{R}_x(\pi)$ and the position $\mathbf{p} = [x, y, z]^\mathsf{T}$, the homogeneous transform of *any* pose has the closed form

$$\mathbf{T}(x, y, z, \varphi) = \begin{bmatrix} \cos\varphi & \sin\varphi & 0 & x \\ \sin\varphi & -\cos\varphi & 0 & y \\ 0 & 0 & -1 & z \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{6}$$

Comment. In `final_quintic_trajectories.m` each transform is obtained as $\mathbf{T}_k = \texttt{scara.fkine}(\texttt{ik}(\mathbf{p}_k))$, i.e. the pose is first mapped to joint space with (5) and then back to Cartesian space with the toolbox forward kinematics. This guarantees that the reported transforms are *exactly* the ones the simulated robot realises; (6) is the analytic value they reproduce.

Geometric derivation of the operating heights

All vertical task heights follow directly from the part dimensions of Fig. 1 and the gripper geometry, rather than being chosen ad hoc. From Fig. 1, Part A consists of a body of height $h_A = 0.10$ and a lower (hatched) peg of length $p = 0.08$ and width 0.15, for a total height $H_A = 0.18$; Part B is a block of height $h_B = 0.10$ containing a socket of depth $s = 0.08$ and width 0.15. The matching widths ($0.15 = 0.15$) confirm that the peg of A seats into the socket of B, and the matching axial lengths ($p = s = 0.08$) mean a fully inserted peg bottoms out exactly when A's lower face meets B's rim.

Pick side (Conveyor A, surface $z = 1.50$). Part A rests on belt A with its peg pointing downward, so the peg tip lies on the belt at $z = 1.50$, the body spans [1.58, 1.68], and the top of A is at $z = 1.68$ — exactly the clearance reference used later. The two-finger gripper (palm offset $\ell_g = 0.10$, finger length $\ell_f = 0.20$, TCP at mid-finger) grasps the body, so the TCP is placed at the body mid-height,

$$z_{\text{grasp}} = \tfrac{1}{2}\,(1.58 + 1.68) = 1.63 = 1.50 + 0.13 \text{ m},$$

i.e. 0.13 m above the belt, with the fingertips (0.10 below the TCP) straddling the body. This fixes the constant vertical offset between the TCP and the peg tip, $c = z_{\text{grasp}} - 1.50 = 0.13$ m, which is preserved throughout transport since there is no relative motion once grasped.

Place side (Conveyor B, surface $z = 1.00$). Part B occupies [1.00, 1.10], so its rim (top face) is at $z = 1.10$ (the second clearance reference, "top of Part B") and its socket bottom is at $1.10 - s = 1.02$. Full assembly requires the peg tip to reach the socket bottom while A's lower face seats on the rim; with the fixed offset c this maps to the TCP height

$$z_{\text{assembled}} = 1.02 + c = 1.02 + 0.13 = 1.15 \text{ m},$$

which is precisely P_4. The descent is split for control: the fast phase (5a) lowers the TCP to $z_{\text{rim}} = 1.21$ m, at which the peg has just crossed the rim ($z = 1.10$, reached at TCP 1.23) into

the socket mouth, and the slow phase (5b) performs the remaining controlled insertion together with the 90° screw,

$$\Delta z_{\text{insert}} = z_{\text{rim}} - z_{\text{assembled}} = 1.21 - 1.15 = 0.06 \text{ m},$$

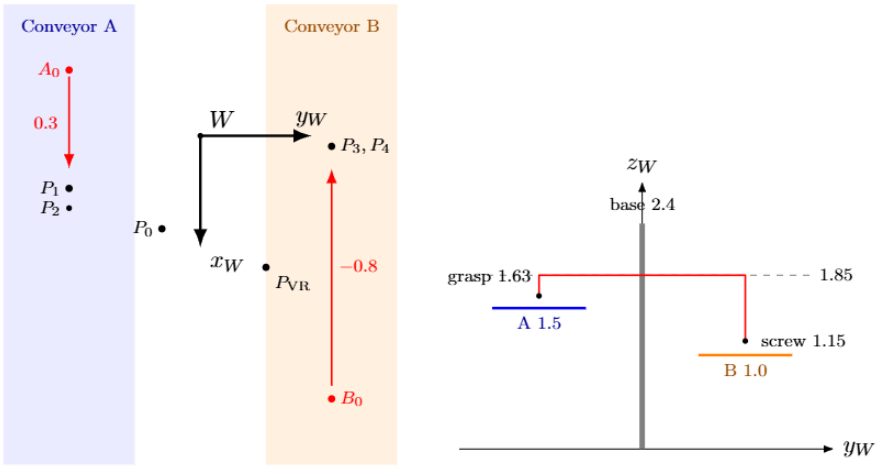
so that the total engagement equals the peg length, $0.02 + 0.06 = 0.08 = p$, as required.

Clearances. Horizontal motion is executed at the transport height $z = 1.85$ m, where the peg tip sits at $1.85 - c = 1.72$ m, clearing both the top of Part A (1.68) and the rim of Part B (1.10). The constant tool offset $0.20 = \ell_g + \frac{1}{2}\ell_f$ is already embedded in T_{tool}, so all of the above heights are expressed consistently at the TCP. Thus the grasp height 1.63, the seating height 1.15, the rim split at 1.21, and the 0.06 m controlled insertion are all consequences of the Fig. 1 geometry and the gripper offset, not free parameters.

The complete set of poses traversed in one assembly cycle, together with their joint coordinates from (5), is listed in Table 1.

Table 1: Intermediate poses (task coordinates) and corresponding joint values. Angles in rad, lengths in m.

Pose	x	y	z	φ	q_1	q_2	q_3	q_4
P_0 (home)	0.707	-0.293	1.85	0.7854	-1.5708	2.3562	0.10	0.0000
P_1 (grasp start)	0.190	-1.000	1.63	0.0000	-2.4199	2.0737	0.32	-0.3462
P_2 (grasp end)	0.340	-1.000	1.63	0.0000	-2.2575	2.0288	0.32	-0.2286
P_{VIA1} (lift)	0.700	-1.000	1.85	0.0000	-1.8744	1.8286	0.10	-0.0457
P_3 (over B)	0.080	1.000	1.85	0.0000	0.4456	2.0907	0.10	2.5363
P_4 (assembled)	0.080	1.000	1.15	1.5708	0.4456	2.0907	0.80	0.9655
P_5 (= P_4)	0.080	1.000	1.15	1.5708	0.4456	2.0907	0.80	0.9655
P_{RETRACT}	0.080	1.000	1.85	1.5708	0.4456	2.0907	0.10	0.9655
$P_{\text{VIA_RET}}$	1.000	0.500	1.85	1.1781	-0.5139	1.9552	0.10	0.2631



As worked examples, the closed form (6) evaluates to

$$\mathbf{T}(P_0) = \begin{bmatrix} 0.7071 & 0.7071 & 0 & 0.707 \\ 0.7071 & -0.7071 & 0 & -0.293 \\ 0 & 0 & -1 & 1.85 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T}(P_1) = \begin{bmatrix} 1 & 0 & 0 & 0.19 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & -1 & 1.63 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T}(P_2) = \begin{bmatrix} 1 & 0 & 0 & 0.34 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & -1 & 1.63 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T}(P_{\text{VIA1}}) = \begin{bmatrix} 1 & 0 & 0 & 0.70 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & -1 & 1.85 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T}(P_3) = \begin{bmatrix} 1 & 0 & 0 & 0.08 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1.85 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T}(P_4) = \begin{bmatrix} 0 & 1 & 0 & 0.08 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1.15 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T}(P_5) = \begin{bmatrix} 0 & 1 & 0 & 0.08 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1.15 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T}(P_{\text{RETRACT}}) = \begin{bmatrix} 0 & 1 & 0 & 0.08 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1.85 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T}(P_{\text{VIA_RET}}) = \begin{bmatrix} 0.3827 & 0.9239 & 0 & 1 \\ 0.9239 & -0.3827 & 0 & 0.5 \\ 0 & 0 & -1 & 1.85 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Why each pose was chosen

P_0 – **home / rest.**
This is the supplied configuration $\boldsymbol{q}_0 = [-90°, 135°, 0.1, 0]$ mapped to task space; the cycle must start and end here. Its orientation $\varphi = 45°$ is *forced* by q_0 through (4) ($-90 + 135 - 0 = 45°$). Geometrically it is a neutral waiting pose at height 1.85, between the two conveyors.

P_1 – **grasp start.**
Directly over Part A at the grasp instant $t = 2.3\,\text{s}$, when A has advanced to $x_A(2.3) = -0.5 + 0.3 \cdot 2.3 = 0.19$. $y = -1$ is the Conveyor A line; $z = 1.63$ places the open fingers around the part (0.13 above the 1.5 belt surface); $\varphi = 0$ aligns the gripper with the part. The x-trajectory reaches this pose with terminal velocity $0.3\,\text{m/s}$ to match the belt.

P_2 – **grasp end.**
The part position after the $0.5\,\text{s}$ closing interval, during which the robot travels at belt speed: $x = 0.19 + 0.3 \cdot 0.5 = 0.34$. Same height and orientation as P_1; only x advances. It marks the instant the gripper is closed and the robot takes over the load.

P_{VIA1} – **lift via point (extra).**
Vertically above P_2 but raised to the transport height 1.85. Pulling the lift point outboard to $x = 0.70$ keeps the arm in a well-conditioned posture, away from both the outer reach limit (2 m) and the inner folded singularity ($q_2 \to 180°$); it is less folded than lifting straight up at the grasp point ($x = 0.34$, $q_2 \approx 116°$), giving the wrist more margin before the sweep across the base.

P_3 – **over Part B.**
Directly above the assembly station. Part B stops at $x_B(2.4) = 2.0 - 0.8 \cdot 2.4 = 0.08$, $y = 1.0$; the transfer ends here at transport height with φ still 0° (the screw rotation comes next).

P_4 – **assembled.**
The screwing pose: the part is lowered onto B ($z = 1.15$) and rotated by 90° ($\varphi : 0 \to 90°$), the rotation the task requires.

$P_5 = P_4$ – **release hold.**
Identical pose, held for the 0.5 s the gripper needs to open.

P_{RETRACT} – **retract (extra).**
Straight-up lift from P_4 to 1.85 after release, keeping $\varphi = 90°$; it takes the tool above all obstacles before the return.

$P_{\text{VIA_RET}}$ – **return fly-by (extra).**
A waypoint at transport height that bends the homeward path into a smooth arc instead of a straight reversal. Its orientation 67.5° lies between 90° (P_4) and 45° (P_0), so φ interpolates monotonically; its position $(1.0, 0.5)$ pulls the path outward to avoid a cusp.

Requirement 2 — Cartesian trajectory design

Building block: quintic polynomial with full boundary conditions

Every smooth segment of every coordinate is generated by a single quintic

$$s(\tau) = \sum_{k=0}^{5} k_k \, \tau^k, \qquad \tau = t - t_0 \in [0, H], \quad H = t_f - t_0, \tag{7}$$

with the six boundary conditions $s(0) = q_0$, $\dot{s}(0) = v_0$, $\ddot{s}(0) = a_0$ and $s(H) = q_f$, $\dot{s}(H) = v_f$, $\ddot{s}(H) = a_f$. Solving the linear system yields

$$\begin{aligned} &k_0 = q_0, \quad k_1 = v_0, \quad k_2 = \tfrac{1}{2}a_0, \\ &A = q_f - q_0 - v_0 H - \tfrac{1}{2}a_0 H^2, \quad B = v_f - v_0 - a_0 H, \quad C = a_f - a_0, \\ &k_3 = \frac{10A - 4BH + \frac{1}{2}CH^2}{H^3}, \quad k_4 = \frac{-15A + 7BH - CH^2}{H^4}, \quad k_5 = \frac{6A - 3BH + \frac{1}{2}CH^2}{H^5}. \end{aligned} \tag{8}$$

Comment (why degree 5). A fifth-order polynomial is the minimal degree that allows prescribing position, velocity *and* acceleration at both ends. Imposing these gives C^2 continuity at every junction, so the end-effector motion has continuous velocity (as required) and, in addition, continuous, bounded acceleration — avoiding the velocity jumps and acceleration impulses that lower-order (e.g. cubic or trapezoidal) blends would introduce between segments.

Scene kinematics and synchronisation constraints

Conveyor A (height 1.5) carries Part A at constant velocity 0.3 m/s along $+x$; from the layout of Fig. 2 its position is

$$x_A(t) = -0.5 + 0.3\,t, \qquad y_A = -1, \; z_A = 1.5. \tag{9}$$

Conveyor B (height 1.0) performs a single step of velocity $-0.8\,\mathrm{m/s}$ for 2.4 s; hence Part B moves along $-x$ and then stops:

$$x_B(t) = \begin{cases} 2.0 - 0.8\,t, & 0 \le t \le 2.4 \\ 0.08, & t > 2.4 \end{cases} \qquad y_B = 1.0,\ z_B = 1.0. \tag{10}$$

Note $x_B(2.4) = 2.0 - 1.92 = 0.08$, which fixes the assembly station at $(x, y) = (0.08, 1.0)$ — exactly the planar coordinates of P_3, P_4.

Catching the moving Part A. To grasp without relative slip, the TCP must match both the position and the velocity of Part A at the grasp instant. With the approach segment ending at $t = 2.3$,

$$x_{\mathrm{TCP}}(2.3) = x_A(2.3) = 0.19, \qquad \dot{x}_{\mathrm{TCP}}(2.3) = \dot{x}_A = 0.3\ \mathrm{m/s}. \tag{11}$$

The first condition fixes $P_1 = (0.19, -1, 1.63, 0)$; the second is imposed through the terminal-velocity boundary condition $v_f = 0.3$ on the x-quintic of Segment 1. During Segment 2 the TCP then translates at the *same* constant $0.3\,\mathrm{m/s}$ as the belt, so the relative pose stays constant for the 0.5 s the gripper needs to close (Step 3 of the task).

Assembling on the stationary Part B. Part B is already at rest (for $t > 2.4$) long before the robot reaches the station at $t = 7$; the descent–screw motion occupies $t \in [7, 8.75]$, so the assembly is performed while Part B is motionless, as required.

Segment decomposition, durations and boundary conditions

The 12 s cycle is split into seven phases (Table 2); the assembly is realised in two sub-phases (5a/5b) and the return in three (7a/7b/7c), for ten polynomial/linear segments in total. Only the coordinates that move in a phase use a quintic; the others are held constant.

Table 2: Trajectory segments. "→" denotes a quintic (8); "const" a held value; "lin" a constant-velocity ramp.

#	Phase	$[t_0, t_f]$ (s)	Coordinate behaviour / boundary conditions
1	Approach $P_0 \to P_1$	$[0, 2.3]$	$x : 0.707 \to 0.19,\ v_f = 0.3$; y, φ quintic $v = a = 0$; z held at 1.85 on $[0, 1.3]$, then quintic $1.85 \to 1.63$ on $[1.3, 2.3]$
2	Grasp follow $P_1 \to P_2$	$[2.3, 2.8]$	x lin. $0.19 + 0.3(t - 2.3)$ ($v = 0.3$ const); y, z, φ const
3	Lift $P_2 \to P_{\mathrm{VIA1}}$	$[2.8, 3.6]$	$x : 0.34 \to 0.70,\ v_0 = 0.3, v_f = 0$; $z : 1.63 \to 1.85$; y, φ const
4	Transfer $P_{\mathrm{VIA1}} \to P_3$	$[3.6, 7.0]$	x, y quintic $v = a = 0$ at both ends, at safe height $z = 1.85$, $\varphi = 0$
5a	Descend to rim	$[7.0, 7.8]$	$z : 1.85 \to 1.21$ (z_{rim}); x, y, φ const
5b	Insert + screw $\to P_4$	$[7.8, 8.75]$	$z : 1.21 \to 1.15$; $\varphi : 0 \to \pi/2$ (90° screw); x, y const
6	Release pause $P_4 \to P_5$	$[8.75, 9.25]$	all const (gripper opens, 0.5 s)
7a	Retract $P_5 \to P_{\mathrm{RETRACT}}$	$[9.25, 10.05]$	$z : 1.15 \to 1.85$; others const
7b	Fly-by $P_{\mathrm{RETRACT}} \to P_{\mathrm{VIA_RET}}$	$[10.05, 11.15]$	quintic with terminal velocity $v_f = v_{\mathrm{via}}$ (Eq. 13)
7c	Return $P_{\mathrm{VIA_RET}} \to P_0$	$[11.15, 12.0]$	quintic with initial velocity $v_0 = v_{\mathrm{via}}$

Comment (number of segments). Seven task phases are used because the task has seven kinematically distinct sub-goals (approach, grasp, lift, transfer, assemble, release, return). Splitting

on these boundaries is what lets us impose phase-specific boundary conditions — most importantly the velocity match $v_f = 0.3$ at the grasp and the via-point velocities on the return. The assembly is further split into a fast descent to the socket rim (5a) and a slow combined insertion/screw (5b), and the return into a straight retraction (7a) and a non-stopping fly-by (7b–7c); a single global polynomial could not satisfy these simultaneously.

Comment (durations). The two 0.5 s holds (grasp/release) are fixed by the statement. The approach lasts 2.3 s: that is the time for Part A to advance from $x = -0.5$ to the chosen interception point $x = 0.19$ at 0.3 m/s, so the robot can meet the part with matched position and velocity at $t = 2.3$. The remaining time is allocated as 0.8 (lift) + 3.4 (transfer) + 0.8 + 0.95 (assembly) + 0.8 (retract) + 1.1 + 0.85 (return). The transfer (Phase 4) is given the largest share (3.4 s) because it covers the largest displacement ($\Delta y = 2$ m) and a slow traverse keeps the end-effector speed moderate; $0.8 + 0.95 + 0.8 + 1.1 + 0.85 = 12$ closes the cycle exactly back at P_0.

Comment (intermediate via points). Three extra poses beyond the task-defined ones are introduced:

- P_{VIA1} raises the part to the safe transport height $z = 1.85$ *before* the long horizontal move, decoupling the vertical lift from the horizontal transfer.
- P_{RETRACT} lifts the tool straight up after release, so the return path starts above all obstacles.
- $P_{\text{VIA_RET}} = (1.0, 0.5, 1.85, 3\pi/8)$ shapes a smooth, non-stopping return curve to P_0. Its orientation $3\pi/8$ lies between $\pi/2$ (at P_4) and $\pi/4$ (at P_0), so φ interpolates monotonically on the way home.

Fly-by via-point velocity

To avoid a full stop at $P_{\text{VIA_RET}}$, the segment is traversed with a non-zero transit velocity. For each coordinate j, let

$$V_{\text{prev}} = \frac{P_{\text{VIA_RET},j} - P_{\text{RETRACT},j}}{T_{7b}}, \qquad V_{\text{next}} = \frac{P_{0,j} - P_{\text{VIA_RET},j}}{T_{7c}}, \tag{12}$$

with $T_{7b} = 1.1$ s and $T_{7c} = 0.85$ s the average-velocity horizons into and out of the via point. The transit velocity is their sign-matched geometric mean,

$$v_{\text{via},j} = \begin{cases} \operatorname{sign}(V_{\text{prev}})\sqrt{|V_{\text{prev}}\, V_{\text{next}}|}, & \operatorname{sign}(V_{\text{prev}}) = \operatorname{sign}(V_{\text{next}}), \\ 0, & \text{otherwise.} \end{cases} \tag{13}$$

Comment. Using the geometric mean of the surrounding average velocities produces a velocity consistent with the local motion on both sides, so the path bends smoothly through $P_{\text{VIA_RET}}$ without halting. When a coordinate reverses direction at the via point (opposite signs), a non-zero transit velocity would overshoot, so it is set to zero for that coordinate, recovering a clean turning point. Because the velocity is shared as v_f of Segment 7b and v_0 of Segment 7c, the velocity remains continuous across the via point.

Degree 5 is the minimum that lets us prescribe position, velocity *and* acceleration at both ends, giving C^2 continuity (hence continuous velocity and bounded, continuous acceleration) at every junction.

Segment-by-segment trajectories with explicit polynomials

Below, constant coordinates are stated; coefficients are rounded to 4 decimals and τ is measured from each segment start.

Phase 1 — Approach $P_0 \to P_1$, $t \in [0, 2.3]$ ($\tau = t$, H=2.3). x ends with the belt velocity ($v_f = 0.3$); y, φ are rest-to-rest; z is held at 1.85 for $t \in [0, 1.3]$ and descends only on $[1.3, 2.3]$ ($\tau' = t - 1.3$), so the approach is flown high and the short descent happens just over Part A.

$$\begin{aligned}
x_1(\tau) &= 0.7070 - 0.6518\,\tau^3 + 0.4497\,\tau^4 - 0.0804\,\tau^5,\\
y_1(\tau) &= -0.2930 - 0.5811\,\tau^3 + 0.3790\,\tau^4 - 0.0659\,\tau^5,\\
\varphi_1(\tau) &= 0.7854 - 0.6455\,\tau^3 + 0.4210\,\tau^4 - 0.0732\,\tau^5,\\
z_1(\tau') &= 1.8500 - 2.2000\,\tau'^3 + 3.3000\,\tau'^4 - 1.3200\,\tau'^5, \quad \tau' \in [0, 1].
\end{aligned}$$

Phase 2 — Grasp follow $P_1 \to P_2$, $t \in [2.3, 2.8]$. Pure belt-following (constant velocity, zero relative motion):

$$x_2(t) = 0.19 + 0.3\,(t - 2.3), \qquad y_2 = -1,\ z_2 = 1.63,\ \varphi_2 = 0.$$

Phase 3 — Lift $P_2 \to P_{\text{VIA1}}$, $t \in [2.8, 3.6]$ ($\tau = t - 2.8$, H=0.8). x decelerates from 0.3 to 0 while z rises to transport height; y, φ constant.

$$\begin{aligned}
x_3(\tau) &= 0.3400 + 0.3000\,\tau + 4.2187\,\tau^3 - 8.4961\,\tau^4 + 4.3945\,\tau^5,\\
z_3(\tau) &= 1.6300 + 4.2969\,\tau^3 - 8.0566\,\tau^4 + 4.0283\,\tau^5, \qquad y_3 = -1,\ \varphi_3 = 0.
\end{aligned}$$

Phase 4 — Transfer $P_{\text{VIA1}} \to P_3$, $t \in [3.6, 7.0]$ ($\tau = t - 3.6$, H=3.4). The long horizontal move at the safe height $z = 1.85$; both x and y are rest-to-rest, z, φ constant.

$$\begin{aligned}
x_4(\tau) &= 0.7000 - 0.1577\,\tau^3 + 0.0696\,\tau^4 - 0.0082\,\tau^5,\\
y_4(\tau) &= -1.0000 + 0.5089\,\tau^3 - 0.2245\,\tau^4 + 0.0264\,\tau^5, \qquad z_4 = 1.85,\ \varphi_4 = 0.
\end{aligned}$$

Phase 5a — Descend to rim, $t \in [7.0, 7.8]$ ($\tau = t - 7.0$, H=0.8). Fast vertical approach to the socket rim $z_{\text{rim}} = 1.21$; x, y, φ constant.

$$z_{5a}(\tau) = 1.8500 - 12.5000\,\tau^3 + 23.4375\,\tau^4 - 11.7188\,\tau^5, \qquad x{=}0.08,\ y{=}1,\ \varphi{=}0.$$

Phase 5b — Insert + screw $\to P_4$, $t \in [7.8, 8.75]$ ($\tau = t - 7.8$, H=0.95). Final 0.06 m insertion together with the 90° screw; x, y constant.

$$\begin{aligned}
z_{5b}(\tau) &= 1.2100 - 0.6998\,\tau^3 + 1.1050\,\tau^4 - 0.4652\,\tau^5,\\
\varphi_{5b}(\tau) &= 18.3210\,\tau^3 - 28.9279\,\tau^4 + 12.1802\,\tau^5, \qquad x_{5b} = 0.08,\ y_{5b} = 1.
\end{aligned}$$

(φ_{5b} runs $0 \to \pi/2$ rad; equivalently $0 \to 90°$.)

Phase 6 — Release pause $P_4 \to P_5$, $t \in [8.75, 9.25]$. All coordinates held at P_4 while the gripper opens.

Phase 7a — Retract $P_5 \to P_{\text{RETRACT}}$, $t \in [9.25, 10.05]$ ($\tau = t - 9.25$, H=0.8). Straight-up lift; x, y, φ constant.

$$z_{7a}(\tau) = 1.1500 + 13.6719\,\tau^3 - 25.6348\,\tau^4 + 12.8174\,\tau^5.$$

Phase 7b/7c — Return via fly-by point. With $T_{7b} = 1.1$, $T_{7c} = 0.85$, Eq. (13) gives numerically

$$v_{\text{via}} = (0,\ -0.6512,\ 0,\ -0.4061) \quad \text{for } (x, y, z, \varphi).$$

The x-component is 0 because x *reverses* at the via point ($V_{\text{prev}} = +0.84 > 0$, $V_{\text{next}} = -0.34 < 0$): a turning point must have zero velocity, otherwise it would overshoot. The y and φ motions are monotonic (both V negative), so they pass through with finite velocity.

Phase 7b $P_{\text{RETRACT}} \to P_{\text{VIA_RET}}$, $t \in [10.05, 11.15]$ ($\tau = t - 10.05$, H=1.1), $v_f = v_{\text{via}}$:

$$\begin{aligned} x_{7b}(\tau) &= 0.0800 + 6.9121\,\tau^3 - 9.4256\,\tau^4 + 3.4275\,\tau^5, \\ y_{7b}(\tau) &= 1.0000 - 1.6038\,\tau^3 + 1.6978\,\tau^4 - 0.5284\,\tau^5, \\ \varphi_{7b}(\tau) &= 1.5708 - 1.6079\,\tau^3 + 1.8874\,\tau^4 - 0.6309\,\tau^5, \qquad z_{7b} = 1.85. \end{aligned}$$

Phase 7c $P_{\text{VIA_RET}} \to P_0$, $t \in [11.15, 12.0]$ ($\tau = t - 11.15$, H=0.85), $v_0 = v_{\text{via}}$:

$$\begin{aligned} x_{7c}(\tau) &= 1.0000 - 4.7710\,\tau^3 + 8.4194\,\tau^4 - 3.9621\,\tau^5, \\ y_{7c}(\tau) &= 0.5000 - 0.6512\,\tau - 7.5048\,\tau^3 + 14.3041\,\tau^4 - 6.9808\,\tau^5, \\ \varphi_{7c}(\tau) &= 1.1781 - 0.4061\,\tau - 3.0218\,\tau^3 + 5.9939\,\tau^4 - 2.9763\,\tau^5, \qquad z_{7c} = 1.85. \end{aligned}$$

(The φ, y values here are in rad; 1.5708=$\frac{\pi}{2}$, 1.1781=$\frac{3\pi}{8}$, 0.7854=$\frac{\pi}{4}$.)

Collision avoidance

All long horizontal motions (Phase 4 transfer and the Phase 7 return) are executed at the upper level $z = 1.85$, which clears Conveyor A (top of Part A at $z \approx 1.68$) and Conveyor B (top of Part B at $z \approx 1.1$). The tool descends only at the two stations: in Phase 1 the descent to the grasp height $z = 1.63$ is delayed to $t \geq 1.3$ (the first 1.3 s are flown at 1.85), and in Phases 5a/5b the descent to $z = 1.15$ happens directly over Part B. Conversely, the tool is lifted back to 1.85 (Phases 3 and 7a) before any horizontal travel. This "lift–traverse–lower" policy keeps the manipulator clear of the scene geometry throughout the cycle.

Continuity summary

By construction the boundary conditions chain across phases: v_x is continuous through the grasp ($0 \to 0.3 \to 0.3 \to 0$ over Phases 1–3), all transfer/assembly endpoints have zero velocity and acceleration, the two holds have zero velocity, and the return reuses the shared v_{via}. The resulting Cartesian and joint trajectories are therefore C^1 (continuous velocity) everywhere — and C^2 wherever the prescribed end accelerations are zero — satisfying the "continuous motion with continuous velocity" requirement.

Number of distinct trajectories. The end-effector motion is decomposed into seven task phases, realised by ten polynomial/linear segments (assembly split into 5a–5b, return into 7a–7c). This is the *minimum* decomposition that can satisfy all task constraints simultaneously, because each boundary exists to impose a condition a single global polynomial could not: (i) the grasp requires a dedicated constant-velocity "follow" segment (Phase 2) so the tool matches the belt velocity exactly and keeps zero relative motion while the gripper closes; (ii) collision avoidance requires separating the vertical lift/lower from the horizontal transfer (Phases 3, 5a, 7a); (iii) the non-stopping return requires splitting Phase 7 so a finite via-point velocity can be shared between the incoming and outgoing pieces. A coarser decomposition would violate at least one of the synchronisation, collision or continuity requirements; a finer one would add segments with no extra task meaning.

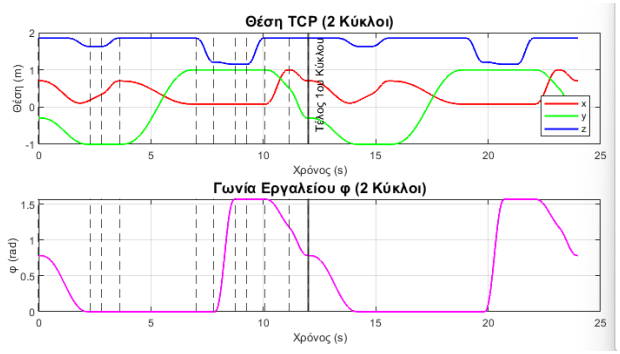
Duration of each trajectory. The total is fixed at 12 s. Two durations are not free: the 0.5 s grasp and 0.5 s release are imposed by the gripper actuation time. The 2.3 s approach is set by the synchronisation: Part A travels from $x = -0.5$ to the interception point $x = 0.19$ (a displacement of 0.69 m) at 0.3 m/s, which takes exactly 2.3 s, so the robot meets the part at $t = 2.3$. The remaining time is distributed as 0.8 (lift) + 3.4 (transfer) + 0.8 + 0.95 (assembly) + 0.8 (retract) + 1.1+0.85 (return): the transfer receives the largest share because it covers the largest displacement ($\Delta y = 2$ m) and a longer time keeps the peak Cartesian velocity and acceleration moderate; the assembly is split into a 0.8 s descent to the rim and a 0.95 s slow insertion/screw. The allocation $2.3 + 0.5 + 0.8 + 3.4 + 0.8 + 0.95 + 0.5 + 0.8 + 1.1 + 0.85 = 12$ s brings the tool back to P_0 exactly at the start of the next cycle.

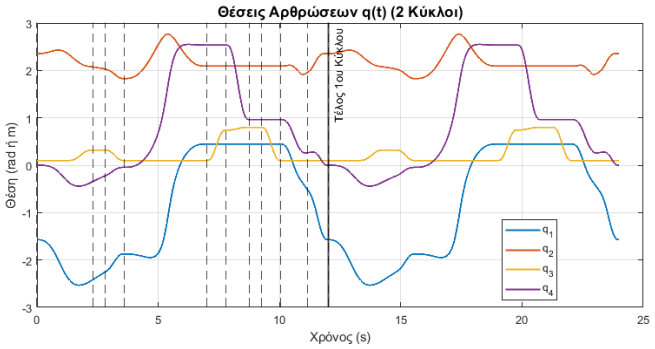
Additional intermediate positions. Beyond the task-defined poses, three via points are introduced, each justified by geometry rather than by the task itself. P_{VIA1} lifts the grasped part to the safe transport height $z = 1.85$ before the horizontal transfer, decoupling the vertical and horizontal motions and letting the longitudinal velocity settle to zero; without it the part would sweep across the scene at grasp height and collide with the conveyors. P_{RETRACT} performs the symmetric straight-up retraction after release, clearing the assembly station before any horizontal travel. $P_{\text{VIA_RET}}$ shapes the homeward path into a smooth, non-stopping arc: its position $(1.0, 0.5)$ pulls the path outward to avoid a cusp, and its orientation 67.5° lies midway between 90° (P_4) and 45° (P_0), so φ interpolates monotonically back to the home orientation. Together these three points enforce the "lift–traverse–lower" policy that keeps the manipulator clear of the conveyors and of Parts A and B throughout the cycle.

Requirement 3 – Simulation Results

Position of end effector and joints

Two consecutive cycles are shown. Every signal has a continuous slope at all phase boundaries (continuous velocity, C^1), and the second cycle reproduces the first exactly — the tool returns precisely to P_0 with zero velocity — confirming the cyclic design.





TCP height z. Held at the transport level 1.85 m except for two descents: a shallow one to the grasp height 1.63 m over $[1.3, 2.3]$ ($\Delta z = 0.22$ m), and a deep one to the assembly height 1.15 m over $[7.0, 8.75]$ — most of it ($1.85 \to 1.21$) in the $[7.0, 7.8]$ approach to the rim, then the final 0.06 m insertion over $[7.8, 8.75]$. Both descents have flat tops because z is frozen during all horizontal motion.

TCP x. During the approach x goes $0.707 \to 0.19$, with a visible undershoot to ≈ 0.107 m at $t \approx 1.86$ s before climbing back to 0.19 m at $t = 2.3$ with slope +0.3 m/s. This undershoot is

unavoidable: the prescribed terminal velocity (+0.3, to match the belt) has the *opposite* sign to the net displacement (−0.517), so the quintic must pass below the target and return. In $[2.3, 2.8]$ x is a straight line at 0.3 m/s ($0.19 \to 0.34$); in $[2.8, 3.6]$ it rises to 0.70 as the longitudinal velocity decays to zero; in $[3.6, 7.0]$ it sweeps to 0.08; it is held at 0.08 during assembly and retraction, then on the return it bulges out to ≈ 1.0 m ($P_{\text{VIA_RET}}$) before settling back to 0.707 m. That outward bulge is what shapes the smooth fly-by.

TCP y. Monotonic $-0.293 \to -1.0$ in the approach, held at -1.0 through grasp and lift $[2.3, 3.6]$, then the large S-shaped rise $-1.0 \to 1.0$ during the transfer $[3.6, 7.0]$, held at 1.0 during assembly and retraction, and finally $1.0 \to 0.5 \to -0.293$ on the return. The transfer dominates the y signal (the largest single displacement, $\Delta y = 2$ m, which is why 3.4 s were allocated to it).

Tool angle φ. $\pi/4 \to 0$ during the approach (the home orientation relaxes to the grasp orientation), then held at 0 through grasp, lift, transfer and the rim descent $[2.3, 7.8]$: the part keeps its grabbed orientation while it travels. The required 90° screw is concentrated in the 0.95 s window $[7.8, 8.75]$, $\varphi : 0 \to \pi/2$. It is held at $\pi/2$ during retraction and then returns $\pi/2 \to \pi/4$ through the via value $3\pi/8 \approx 1.178$ rad; the slope changes visibly but remains continuous (the fly-by does not stop) and flattens as it reaches $\pi/4$ ($v_f = 0$) at P_0.

Joint q_1. Tracks the planar (x, y) motion: $-\pi/2$ at home, dipping to ≈ -2.42 rad ($-139°$) at the grasp, then sweeping up to $+0.45$ rad through the transfer, held during the assembly (x, y fixed), and returning to $-\pi/2$.

Joint q_2. $3\pi/4$ at home, decreasing to ≈ 1.83 rad by P_{VIA1}, then rising to a maximum ≈ 2.77 rad (158.5°) at $t \approx 5.4$ s in the middle of the transfer. This peak is the elbow at its most folded: there the Cartesian path passes closest to the base ($\min(x^2 + y^2) \approx 0.139$), which requires a larger elbow angle. Importantly q_2 stays below π, so the arm never reaches the folded singularity. It then settles to 2.09 rad at P_3/P_4.

Joint q_3. Exactly $1.95 - z$: flat at 0.1 m at transport level, a small bump to 0.32 m at the grasp, back to 0.1, then the large bump to 0.8 m at the assembly, back to 0.1. The two bumps coincide precisely with the two z-descents.

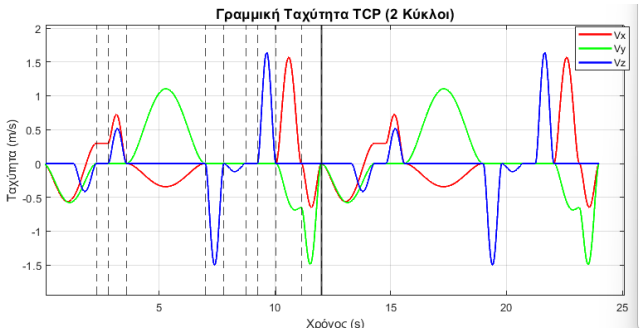
Joint q_4. During the transfer the tool angle φ is held at 0, yet the arm angle $q_1 + q_2$ swings by about 2.5 rad; since $\varphi = q_1 + q_2 - q_4$, the wrist must rotate by the same amount to keep the orientation fixed, so q_4 climbs to ≈ 2.54 rad following $q_1 + q_2$. Then during the assembly the arm is stationary and q_4 *alone* produces the screw: it drops by exactly $\pi/2$ ($2.54 \to 0.97$ rad). Thus q_4 performs a double duty — compensating the arm rotation to preserve orientation during transport, and delivering the 90° rotation during assembly — and it unwinds to 0 (home) on the return. Note that the 90° requirement appears here as a *change* of q_4, not as an absolute value of 90°.

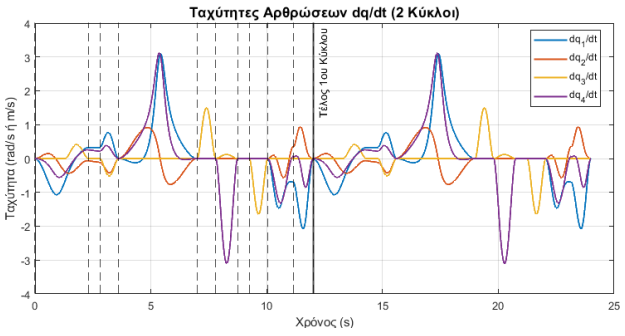
Continuity and synchronisation. All Cartesian and joint signals have continuous slope at the dashed boundaries (continuous velocity). In particular the slope of x at $t = 2.3$ is exactly +0.3 m/s and stays constant through $[2.3, 2.8]$: this is the visible signature of the belt-velocity match and of the zero-relative-motion grasp. At $t = 12$ (solid line) every curve joins the second cycle seamlessly, confirming a clean periodic return to P_0.

Velocity Profiles

Each velocity is the time-derivative of the corresponding position trajectory, so every moving quintic appears as a single bell-shaped pulse. All traces are continuous at the phase boundaries

(no vertical jumps ⇒ continuous velocity), and most also have continuous slope there, because the end accelerations of the quintics were set to zero (⇒ continuous acceleration).





TCP velocity $\dot{z}$. Zero except for the four vertical moves: $-0.41\,\mathrm{m/s}$ at the grasp descent $[1.3, 2.3]$, $+0.52\,\mathrm{m/s}$ at the lift $[2.8, 3.6]$, $-1.50\,\mathrm{m/s}$ at the rim descent $[7.0, 7.8]$, and $+1.64\,\mathrm{m/s}$ at the retraction $[9.25, 10.05]$. The retraction pulse is the tallest because $0.70\,\mathrm{m}$ is covered in only $0.8\,\mathrm{s}$; $\dot{z} = 0$ throughout the horizontal transfer, where z is held.

TCP velocity $\dot{x}$. A small negative lobe in the approach that *rises to $+0.3\,m/s$ exactly at* $t = 2.3$ (the belt speed) and stays flat at $+0.3$ through the $0.5\,\mathrm{s}$ grasp — the constant-velocity, zero-relative-motion follow. It then decays to 0 by $t = 3.6$, shows a $-0.34\,\mathrm{m/s}$ lobe during the transfer, and finally the large return pulses: $+1.57\,\mathrm{m/s}$ outward (fly-by, peak at $t \approx 10.6$) and $-0.65\,\mathrm{m/s}$ back to P_0.

TCP velocity $\dot{y}$. A $-0.58\,\mathrm{m/s}$ lobe in the approach, zero during grasp and lift, the dominant $+1.10\,\mathrm{m/s}$ lobe during the transfer (largest displacement, $\Delta y = 2\,\mathrm{m}$, but kept moderate by the $3.4\,\mathrm{s}$ allocation), zero during assembly and retraction, and then a negative excursion over $[10.05, 12]$ reaching $\approx -1.49\,\mathrm{m/s}$. Its value at the via point ($-0.65\,\mathrm{m/s}$ at $t = 11.15$) is shared continuously between the two return segments: this is the fly-by, which avoids stopping.

Joint velocity $\dot{q}_3$. Exactly $-\dot{z}$ (since $q_3 = 1.95 - z$): a mirror image of the blue trace, with $+1.50$ at the rim descent and -1.64 at the retraction.

Joint velocity $\dot{q}_2$. The smallest of the four, $|\dot{q}_2| \lesssim 1\,\mathrm{rad/s}$, a mild $\pm$ lobe during the transfer and on the return: the elbow angle changes only moderately, so its rate stays low.

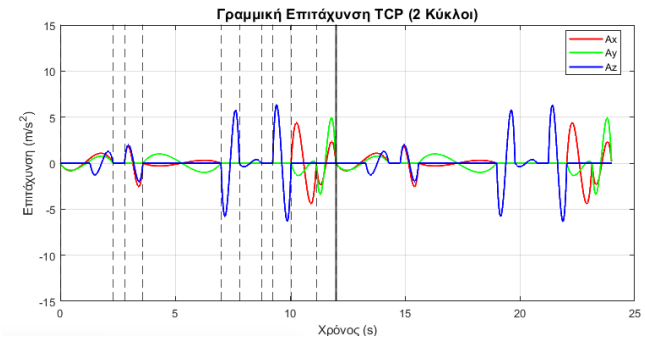
Joint velocities $\dot{q}_1$ and $\dot{q}_4$. Both spike to $\approx +3.1\,\mathrm{rad/s}$ at $t \approx 5.4\,\mathrm{s}$, in the middle of the transfer — far larger than any Cartesian speed there ($\dot{y} \approx 1.10\,\mathrm{m/s}$). This is *Jacobian amplification*: at that instant the arm is at its most folded posture ($q_2 \approx 158.5°$, the path's closest approach to the base, $x^2 + y^2 \approx 0.139$), so the manipulator Jacobian is ill-conditioned and a moderate end-effector velocity maps to large joint rates. The arm never reaches the folded singularity ($q_2 < \pi$), so the rates remain finite but peak there. Because $\dot{q}_4 = \dot{q}_1 + \dot{q}_2 - \dot{\varphi}$ and $\dot{\varphi} = 0$ during the transfer, $\dot{q}_4$ essentially follows $\dot{q}_1$ here.

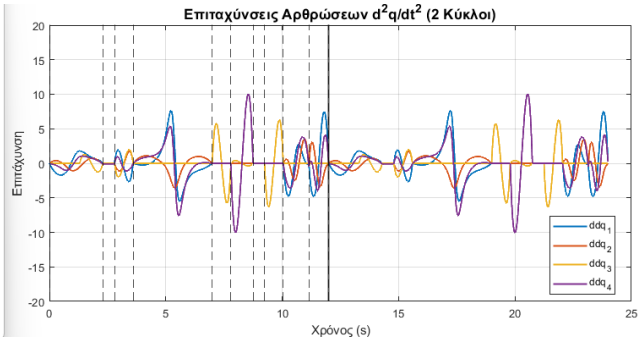
The wrist during assembly. $\dot{q}_4$ then dips to $\approx -3.10\,\mathrm{rad/s}$ at $t \approx 8.28\,\mathrm{s}$. With the arm stationary during the assembly ($\dot{q}_1 = \dot{q}_2 = 0$), one has $\dot{q}_4 = -\dot{\varphi}$, so the wrist alone performs the $90°$ screw at its peak rate ($\approx -3.10\,\mathrm{rad/s}$, the $0 \to \pi/2$ move in $0.95\,\mathrm{s}$). At the same window $\dot{z} \approx -0.12$ (final insertion): the lowering and the screwing happen simultaneously.

Continuity and overall picture. No trace shows a jump (continuous velocity everywhere); at the via point $t = 11.15$ the rates are continuous through the shared v_{via} (fly-by), and all velocities return to zero at $t = 12$ and repeat. Notably, the peak end-effector speed ($\approx 1.64\,\text{m/s}$, retraction) and the peak joint rate ($\approx 3.1\,\text{rad/s}$, folded transfer) both occur in the obstacle-free, non-contact phases, while the delicate grasp and assembly run slowly and smoothly — the schedule deliberately spends the available velocity budget on the free return and keeps the precision phases gentle.

Acceleration Profiles

The acceleration is the second derivative of the position. Each rest-to-rest quintic produces the characteristic *antisymmetric two-lobe* pulse — a positive lobe (accelerate) followed by an equal negative lobe (decelerate), zero at both ends — so every velocity bell of the previous figure appears here as a $+/-$ pair. For a rest-to-rest move of amplitude Δ over time T the peak magnitude is $\approx 5.77\,\Delta/T^2$; the *shortest* moves therefore produce the *largest* accelerations. Crucially, all traces are continuous and return to zero at every boundary: the trajectory is C^2 (continuous acceleration), which exceeds the "continuous velocity" requirement.





TCP acceleration $\ddot{z}$. The dominant component. It shows $\pm 5.77\,\text{m/s}^2$ lobes at the rim descent $[7.0, 7.8]$ ($0.64\,\text{m}$ in $0.8\,\text{s}$) and the largest swings, $\pm 6.31\,\text{m/s}^2$, at the retraction $[9.25, 10.05]$ ($0.70\,\text{m}$ in $0.8\,\text{s}$, $5.77 \cdot 0.7/0.8^2 = 6.31$). It is zero during the horizontal transfer.

TCP acceleration $\ddot{x}$. Small lobes in the approach and during the transfer, then the large $\pm 4.39\,\text{m/s}^2$ pair at the outward fly-by $[10.05, 11.15]$ ($5.77 \cdot 0.92/1.1^2 = 4.39$).

TCP acceleration $\ddot{y}$. Modest throughout: a small pair in the approach, a low-amplitude pair during the $3.4\,\text{s}$ transfer (long time $\Rightarrow$ small acceleration despite the $2\,\text{m}$ displacement), and a $+4.91/-3.38\,\text{m/s}^2$ excursion on the return.

Joint acceleration $\ddot{q}_3$. Exactly $-\ddot{z}$ (since $q_3 = 1.95 - z$): a mirror of the blue TCP trace, with the ± 6.31 swings at the retraction.

Joint acceleration $\ddot{q}_2$. The smallest, $|\ddot{q}_2| \lesssim 3.6$: the elbow contributes little.

Joint acceleration $\ddot{q}_1$. Reaches $+7.65/-5.50\,\text{rad/s}^2$, peaking in the transfer ($t \approx 5.2$–$5.4\,\text{s}$) near the folded posture. As with the velocity, this is Jacobian amplification: a modest Cartesian acceleration maps to a large joint acceleration where the arm is most bent.

Joint acceleration $\ddot{q}_4$. It reaches $\pm 10.05\,\mathrm{rad/s^2}$ at $t \approx 8.55\,\mathrm{s}$. During the assembly the arm is stationary, so $\ddot{q}_4 = -\ddot{\varphi}$, and this value is precisely the angular acceleration of the 90° screw executed in 0.95 s: $5.77 \cdot (\pi/2)/0.95^2 = 10.05\,\mathrm{rad/s^2}$. The wrist thus carries the highest joint acceleration of the whole task, all of it in the brief screwing window.

Continuity and overall picture. No trace shows a jump; every lobe begins and ends at zero, confirming the C^2 property. At the via point $t = 11.15$ the acceleration passes through zero from both sides (a smooth inflection) even though the velocity there is non-zero — the signature of the fly-by. As with the velocities, the largest accelerations ($\ddot{q}_4$ at the screw, $\ddot{z}$ at retraction) fall in the brief, obstacle-free phases, while the long transfer and the contact phases stay gently accelerated — keeping actuator effort and inertial loads on the grasped part low where it matters.

Part 2 – Adaptation with Dynamic Movement Primitives

n Part 2 the part B is no longer placed precisely. At $t = 0$ a camera measures its planar offset $\boldsymbol{\delta} = [\delta_x, \delta_y, 0]^\mathsf{T}$ with $\delta_x \in [-0.1, 0.5]$, $\delta_y \in [-0.2, 0.2]$, and a rotation $\theta_\delta \in [-30°, 30°]$ about z. The factory adapts only the trajectories that depend on part B's pose by means of Dynamic Movement Primitives (DMP) with spatial scaling, reusing the Part 1 trajectories everywhere else. This section gives the DMP formulation and answers Requirement 1: training a *position* DMP from the Part 1 trajectories (the orientation variation is ignored here and treated in Requirement 3).

Trajectory Adaptation

The camera-detected assembly pose becomes

$$\boldsymbol{g}_{P_3} = P_3 + [\delta_x, \delta_y, 0], \qquad \boldsymbol{g}_{P_4} = P_4 + [\delta_x, \delta_y, 0], \qquad \varphi_{\mathrm{goal}} = \tfrac{\pi}{2} + \theta_\delta. \tag{14}$$

Only motions whose goal is part B's pose change; the pick side (part A, conveyor A) is untouched. Hence:

- Segments 1–3 (approach, grasp, lift to P_{VIA1}) are independent of $\boldsymbol{\delta} \Rightarrow$ reused *verbatim* from Part 1.
- The transfer (Segment 4, $P_{\mathrm{VIA1}} \to P_3$) has its goal shifted by $\boldsymbol{\delta} \Rightarrow$ reproduced by a *position DMP* with spatial scaling.
- The assembly descent (Segment 5, z) has $\delta_z = 0$ always $\Rightarrow$ unchanged; its x, y are simply held at the shifted goal $\boldsymbol{g}_{P_3}$.
- The orientation φ is ignored in Requirement 1 (uses the Part 1 quintic $0 \to \pi/2$); a scalar orientation DMP is added in Requirement 3.

Requirement 1 therefore trains a position DMP on the Part 1 transfer trajectory and retargets it to the detected goal by spatial scaling.

DMP formulation

A DMP encodes a demonstrated trajectory as a stable second-order (spring–damper) system attracted to a goal g, augmented by a learnable forcing term that reproduces the demonstrated shape. For one coordinate y with scaled velocity z,

$$\tau\,\dot{z} = \alpha_z\big(\beta_z(g - y) - z\big) + \mathbf{S}_g\, f(x), \qquad \tau\,\dot{y} = z, \tag{15}$$

with positive gains α_z, β_z set to critical damping $\beta_z = \alpha_z/4$. Without f, (15) is a critically damped spring that drives $y \to g$ smoothly. The explicit time dependence of f is removed by the first-order *canonical system* (phase variable x),

$$\tau\,\dot{x} = -\alpha_x x, \quad x(0) = 1 \implies x(t) = \exp\Big(-\tfrac{\alpha_x}{\tau}\,t\Big), \tag{16}$$

which decays monotonically $1 \to 0$, synchronises all degrees of freedom, and makes the forcing vanish at the end. The forcing term is a normalised Gaussian basis expansion modulated by x,

$$f(x) = \frac{\sum_{i=1}^{N} w_i\, \psi_i(x)}{\sum_{i=1}^{N} \psi_i(x)}\, x, \qquad \psi_i(x) = \exp\big(-h_i\,(x - c_i)^2\big), \tag{17}$$

with centres c_i, inverse widths h_i and learnable weights w_i. Since $f \propto x \to 0$, the spring dominates near the goal and convergence to g is guaranteed.

Spatial scaling. To execute the same shape towards a different start/goal, the forcing is pre-multiplied by the diagonal matrix

$$\mathbf{S}_g = \mathrm{diag}(\boldsymbol{g} - \boldsymbol{y}_0)\ \mathrm{diag}(\boldsymbol{g}_d - \boldsymbol{y}_{0d})^{-1}, \tag{18}$$

where $(\boldsymbol{y}_{0d}, \boldsymbol{g}_d)$ are the demonstration's start/goal and $(\boldsymbol{y}_0, \boldsymbol{g})$ the new ones. Each coordinate is scaled by the ratio of its new to its demonstrated displacement, so the trajectory is stretched, shrunk or flipped to begin at $\boldsymbol{y}_0$ and end at $\boldsymbol{g}$ while keeping its shape.

Temporal scaling. τ rescales time: with the demonstration $\tau_d = T_d$, a value τ executes the motion in duration $T = (\tau/\tau_d)T_d$. Here we keep $\tau = \tau_d$, so the adapted transfer lasts the same 3.4 s as in Part 1.

Requirement 1 — Training the position DMP

Demonstration

The desired trajectory is the Part 1 transfer (Segment 4, $P_{\mathrm{VIA1}} \to P_3$): the quintic of duration $T_d = 3.4\,\mathrm{s}$, with $y_d^{(x)} : 0.70 \to 0.08$ and $y_d^{(y)} : -1.0 \to 1.0$. Its derivatives $\dot{y}_d, \ddot{y}_d$ follow from the quintic (analytically or by finite differences). Each coordinate is an independent scalar DMP sharing the single canonical system $x(t)$ of (16).

Forcing term of the demonstration

Set $\tau_d = T_d$, $\boldsymbol{g}_d = \boldsymbol{y}_d(T_d)$, $\mathbf{S}_g = \mathbf{I}$. Substituting $\tau\dot{y} = z$ (so $z = \tau\dot{y}$, $\dot{z} = \tau\ddot{y}$) into (15) and solving for the forcing,

$$\boxed{f_d(t) = \tau_d^2\, \ddot{y}_d(t) - \alpha_z\big(\beta_z(g_d - y_d(t)) - \tau_d\, \dot{y}_d(t)\big).} \tag{19}$$

Canonical system and basis placement

With $x(t) = \exp(-(\alpha_x/\tau_d)t)$, N Gaussians are placed uniformly in time and mapped to the phase:

$$t_i = \frac{i-1}{N-1}\, T_d, \quad c_i = x(t_i) = \exp\Big(-\alpha_x \tfrac{i-1}{N-1}\Big), \quad h_i = \frac{1}{(c_{i+1} - c_i)^2}\ (i < N), \quad h_N = h_{N-1}. \tag{20}$$

α_x is chosen so that $x(T_d)$ is small, i.e. the centres $c_i \in [\,e^{-\alpha_x}, 1\,]$ cover the whole phase range.

Weight learning by Locally Weighted Regression

For each basis the weight minimises the locally weighted squared error between the model $x\,w_i$ and f_d,

$$J_i = \sum_t \psi_i\big(x(t)\big)\, \big(f_d(t) - x(t)\, w_i\big)^2, \tag{21}$$

whose closed-form solution is

$$w_i = \frac{\boldsymbol{F}_d^{\mathsf{T}}\, \boldsymbol{\Gamma}_i\, \boldsymbol{s}}{\boldsymbol{s}^{\mathsf{T}}\, \boldsymbol{\Gamma}_i\, \boldsymbol{s}} = \frac{\sum_t x(t)\, \psi_i(x(t))\, f_d(t)}{\sum_t x(t)^2\, \psi_i(x(t))}, \tag{22}$$

with $\boldsymbol{s} = [x(t_1), \ldots, x(t_M)]^{\mathsf{T}}$, $\boldsymbol{F}_d = [f_d(t_1), \ldots, f_d(t_M)]^{\mathsf{T}}$ and $\boldsymbol{\Gamma}_i = \mathrm{diag}\big(\psi_i(x(t_1)), \ldots\big)$.

Implementation note. In the code the constant factor $1/(g_d - y_{0d})$ is folded into the stored weights and the factor $(g - y_0)$ is applied at roll-out; this is algebraically identical to keeping $\mathbf{S}_g$ explicit as in (18) (verified to machine precision), and reproduces exactly the same generated trajectory.

Execution (roll-out) and retargeting to the camera goal

At run time the camera fixes the new goal — for the x axis $g = 0.08 + \delta_x$, for the y axis $g = 1.0 + \delta_y$ — while the start $\boldsymbol{y}_0 = P_{\text{VIA1}}$ is unchanged. The DMP is integrated forward with Euler and step $\Delta t = 10^{-3}\,\text{s}$ (Hint 2):

$$\begin{aligned} z &\leftarrow z + \tfrac{\Delta t}{\tau}\big[\alpha_z(\beta_z(g - y) - z) + \mathbf{S}_g f(x)\big], \\ y &\leftarrow y + \tfrac{\Delta t}{\tau}\, z, \qquad x \leftarrow x + \tfrac{\Delta t}{\tau}\,(-\alpha_x x), \end{aligned} \tag{23}$$

with the per-coordinate scaling $\mathbf{S}_g = (g - y_0)/(g_d - y_{0d})$. The DMP reproduces the demonstrated, collision-free, smooth transfer shape, retargeted to terminate exactly at the detected part B location.

Parameters

The ratio $\beta_z = \alpha_z/4$ makes the transformation system *critically damped*: the characteristic equation $\tau^2 s^2 + \alpha_z \tau s + \alpha_z \beta_z = 0$ has a double root $s = -\alpha_z/(2\tau)$, the fastest response with no overshoot, so the goal attractor never oscillates around the part—essential when the target is a screw hole. The magnitude $\alpha_z = 25$ then sets the attractor bandwidth: the time constant $2\tau/\alpha_z \approx 0.27\,\text{s}$ is short enough for the trajectory to converge comfortably within the $\tau_d = 3.4\,\text{s}$ horizon, yet small enough that the dynamics stay numerically gentle for explicit Euler integration. The phase gain $\alpha_x = 1.0$ is a standard compromise: it makes the forcing decay to $x(\tau_d) = e^{-1} \approx 0.37$ by the end (so the spring dominates near the goal and ensures convergence) while keeping the basis centres $c_i \in [e^{-1}, 1]$ spread over a useful portion of the phase. $N = 40$ Gaussians are used: already $N \approx 20$ captures this smooth, low-curvature quintic, and $N = 40$ drives the reproduction error to the millimetre level at negligible extra cost while being shared with the orientation DMP. The temporal scale is kept at $\tau = \tau_d = 3.4\,\text{s}$ because the task timing is fixed (12 s per cycle), so no speed-up or slow-down of the demonstration is needed. Finally $\Delta t = 10^{-3}\,\text{s}$ (Hint 2) resolves the fastest mode ($\omega_n = \sqrt{\alpha_z \beta_z} = 12.5\,\text{rad/s}$, period $\approx 0.50\,\text{s}$) with about 500 integration steps per oscillation, giving an accurate and stable roll-out.

Requirement 2 – Simulation results of Positional Adaptation

The three cycles use $\boldsymbol{\delta}_1 = \mathbf{0}$, $\boldsymbol{\delta}_2 = [-0.2, 0.2, 0]$ and $\boldsymbol{\delta}_3 = [0.4, -0.15, 0]$ (the dashed lines at $t = 12, 24\,\text{s}$ mark the cycle boundaries), with the orientation *not* yet adapted ($\theta_\delta = 0$), so only the planar goal of the transfer changes. The detected assembly station moves to $g_{P_3} = (0.08, 1.0)$, $(-0.12, 1.2)$ and $(0.48, 0.85)$ in the three cycles.

A first observation is that everything independent of part B is identical across all cycles, confirming the "reuse Part 1" strategy: the pick side (approach, grasp and lift, the first 3.6 s of each cycle) overlaps exactly; the TCP height z (blue) — and hence the prismatic q_3 (yellow) — is unchanged because $\delta_z = 0$, so the same shallow grasp dip to 1.63 m and deep assembly dip to 1.15 m repeat every cycle; and, since $\theta_\delta = 0$, the tool angle follows the same $0 \to \pi/2$ profile. In particular cycle 1 ($\boldsymbol{\delta} = 0$, spatial scales $S_x = S_y = 1$) reproduces the Part 1 trajectory exactly, which validates the DMP: with no perturbation the learned forcing regenerates the demonstration.

The adaptation is carried entirely by the planar coordinates. The transfer DMP retargets x, y to each shifted goal, so in the TCP plot the red (x) and green (y) curves settle on different plateaus during the assembly hold — x at 0.08, −0.12 and 0.48 m and y at 1.0, 1.2 and 0.85 m, exactly the camera-detected stations. This is the spatial scaling $\mathbf{S}_g = \operatorname{diag}(\boldsymbol{g} - \boldsymbol{y}_0)\operatorname{diag}(\boldsymbol{g}_d - \boldsymbol{y}_{0d})^{-1}$ in action:

the per-coordinate factors are $(S_x, S_y) = (1, 1)$, $(1.32, 1.10)$ and $(0.36, 0.93)$, so in cycle 2 the demonstrated shape is *stretched* (the goal is farther in both axes) while in cycle 3 the x-excursion is *compressed* to 36% (the goal $x = 0.48$ lies much closer to the lift point 0.70). Crucially the *shape* of the path — smooth, monotone, collision-free at the transport height 1.85 m — is preserved in every cycle; only its endpoint moves.

These planar changes propagate into the joints q_1, q_2, q_4 (which depend on x, y) but not into q_3 (which depends only on z). During the assembly hold the plateaus differ accordingly: q_1 sits at 0.45, 0.75, 0.00 rad, q_2 at 2.09, 1.85, 2.12 rad and q_4 at 0.97, 1.02, 0.55 rad in the three cycles, whereas q_3 shows the identical grasp/assembly bumps every time. All three goals lie well inside the workspace ($1 - D^2 \in [0.73, 0.93]$, far from the $q_2 = \pi$ folded singularity), so the adapted motions stay valid and smooth. Finally, every cycle returns exactly to the fixed home P_0 — the return funnels through the fixed via point $P_{\text{VIA_RET}}$ — so the curves join continuously at $t = 12$ and 24 s and the process is cleanly periodic despite the changing targets.

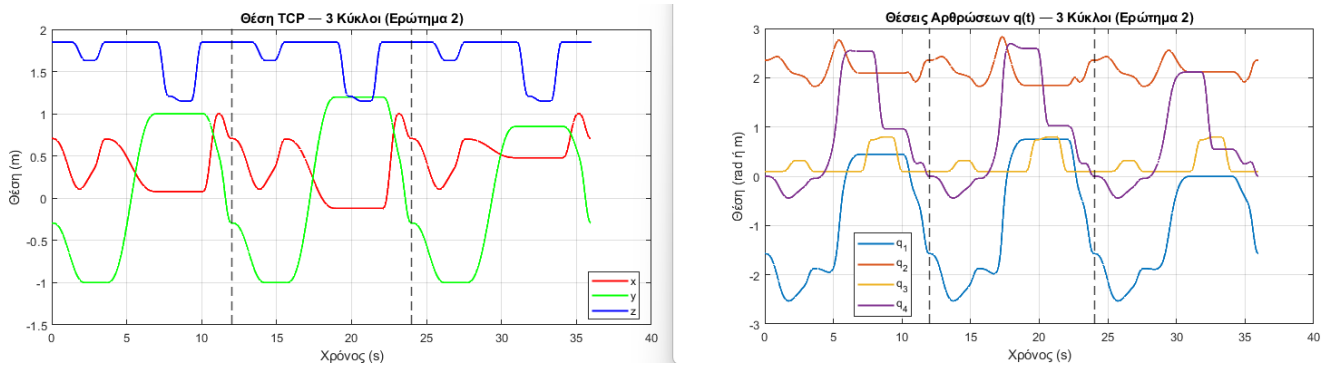
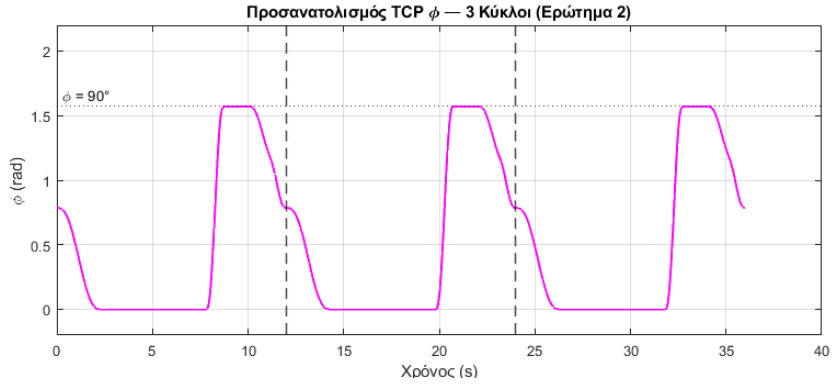


Figure 1: Requirement 2 — TCP Cartesian position (x, y, z) and joint coordinates $q_{1\text{–}4}$ over three cycles for $\boldsymbol{\delta}_1$=0, $\boldsymbol{\delta}_2$=[−0.2, 0.2], $\boldsymbol{\delta}_3$=[0.4, −0.15] ($\theta_\delta = 0$). The pick side and the z/q_3 traces are identical every cycle; only the x, y plateaus during the assembly hold move to the camera-detected station.



Desired and generated DMP trajectories

This is the comparison the statement requires for the DMP-generated motions: in a common plot, the demonstration used for training (dashed black) and the trajectory produced by each of the three simulations, for the two adapted coordinates x (top) and y (bottom) of the transfer phase $P_{\mathrm{VIA1}} \to P_3$, over the 3.4 s horizon.

Reproduction (validation). For $\boldsymbol{\delta}_1 = \mathbf{0}$ the goal is unchanged, so the spatial scales are $S_x = S_y = 1$ and the DMP must regenerate the demonstration. Indeed the red curve lies on top of the dashed desired one in both x and y: the learned forcing term reproduces the training trajectory with negligible error (a residual at the millimetre level, set by the number of basis functions). This confirms that the training of §3 is correct—the DMP is a faithful encoding of the Part 1 transfer before any adaptation.

Spatial scaling. For $\boldsymbol{\delta}_2 = [-0.2, 0.2]$ and $\boldsymbol{\delta}_3 = [0.4, -0.15]$ the goal moves, and the two coloured curves are clean *scaled* versions of the demonstration. They all share the *same start* $(x_0, y_0) = (0.70, -1.0)$ — the lift point P_{VIA1} does not move with $\boldsymbol{\delta}$, only the goal does — and fan out to their respective targets $g_x = 0.08, -0.12, 0.48$ and $g_y = 1.0, 1.2, 0.85$. The direction and amount of the deviation from the desired curve are exactly those predicted by $\mathbf{S}_g = \mathrm{diag}(\boldsymbol{g}-\boldsymbol{y}_0)\,\mathrm{diag}(\boldsymbol{g}_d-\boldsymbol{y}_{0d})^{-1}$: in x, $\boldsymbol{\delta}_2$ has $S_x = 1.32 > 1$ and a lower goal, so its curve runs *below* the desired and descends further, while $\boldsymbol{\delta}_3$ has $S_x = 0.36 < 1$ and a higher goal, so it stays *above* and barely descends; in y, $\boldsymbol{\delta}_2$ ($S_y = 1.10$) rises *above* the desired to 1.2, while $\boldsymbol{\delta}_3$ ($S_y = 0.93$) stays *below* at 0.85. Each axis is scaled by its own factor, independently, because $\mathbf{S}_g$ is diagonal.

Reading the perturbation off the figure. The plot lets one recover $\boldsymbol{\delta}$ *directly*, without touching the code. Every curve starts at the common lift point ($x_0 = 0.70$, $y_0 = -1.0$); only the right-hand endpoint moves. Reading the final plateau of each x curve and subtracting the nominal 0.08 gives $\delta_x = 0$, -0.20, $+0.40$, and the final y minus the nominal 1.0 gives $\delta_y = 0$, $+0.20$, -0.15 — precisely the injected offsets. The spatial scale itself is visible as the *depth of the excursion*: in x the dashed demonstration drops by 0.62 ($0.70 \to 0.08$), whereas $\boldsymbol{\delta}_2$ drops by 0.82 (ratio $0.82/0.62 = 1.32 = S_x$) and $\boldsymbol{\delta}_3$ by only 0.22 (ratio $0.22/0.62 = 0.36 = S_x$); the ratio of any cycle's total drop to the dashed drop reads off S_x at a glance, and likewise S_y from the y panel. The red ($\boldsymbol{\delta}_1$) curve sitting exactly on the dashed line is the visual proof that the DMP reproduces the demonstration when $\boldsymbol{\delta} = 0$ — the validation case — while the way the curves *separate progressively* (coinciding in the first half-second, where $x \approx 1$ and the forcing dominates, then fanning to their goals) is the signature of the goal attractor taking over as the phase decays.

Shape preservation and continuity. Regardless of the goal, every curve keeps the demonstration's smooth, monotone S-shape with *zero slope at both ends*: the rest-to-rest velocity profile of the Part 1 quintic is inherited, so the adapted transfer still leaves P_{VIA1} at rest and arrives at the (shifted) station at rest. This is what guarantees that the DMP segment connects to the preceding lift and the following assembly descent with continuous velocity, exactly as in Part 1. Because the two axes use different factors when $S_x \neq S_y$ (most visibly for $\boldsymbol{\delta}_3$, 0.36 vs 0.93), the *2-D Cartesian path* bends slightly differently between cycles, even though each *per-axis time profile* remains a faithful scaled copy of the demonstration; this mild frame dependence is the known behaviour of the standard diagonal-$\mathbf{S}_g$ DMP and is well within the collision-free corridor here.

Summary The plot demonstrates the two defining DMP properties at once: exact reproduction of the demonstration when $\boldsymbol{\delta} = 0$, and predictable, smooth spatial generalisation to new goals when $\boldsymbol{\delta} \neq 0$ — the part is delivered exactly to the camera-detected station every time, with no re-planning and no loss of the trajectory's smoothness or safety.

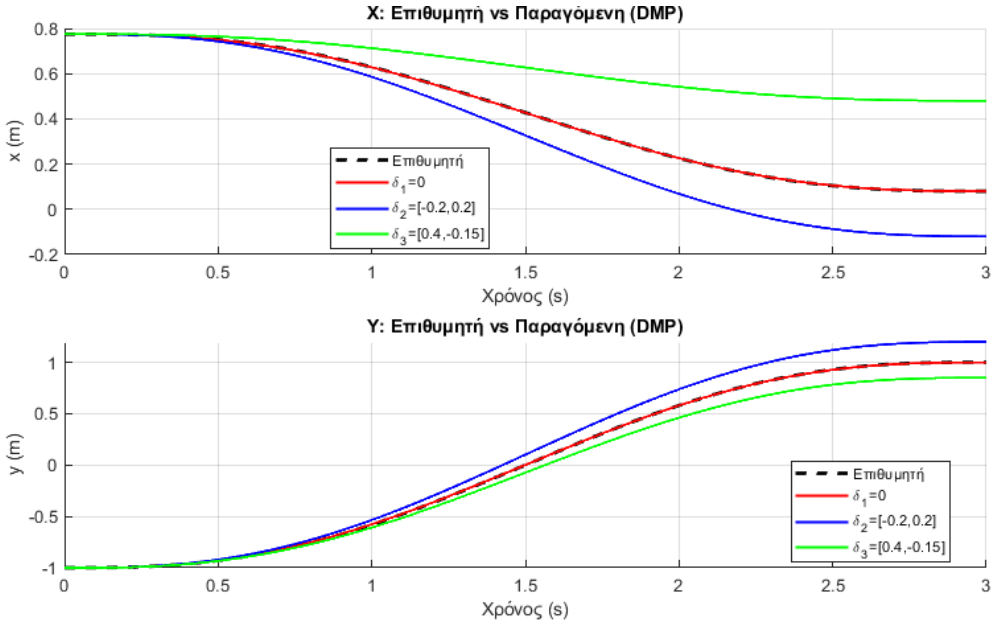


Figure 2: Requirement 2 — Position DMP, desired (dashed black) vs. generated for the three cycles, coordinate x (top) and y (bottom) of the transfer over the 3.4 s horizon. $\boldsymbol{\delta}_1$ (red) lies on the demonstration (validation); $\boldsymbol{\delta}_2, \boldsymbol{\delta}_3$ are scaled copies ending at the detected goals. Reading the endpoints gives $\boldsymbol{\delta}$ and the descent depths give S_x, S_y directly.

Velocity profiles across the three cycles

The velocity traces split cleanly into two families: pulses that *vary* with $\boldsymbol{\delta}$ and pulses that are *identical* in every cycle. This split is the clearest signature that only the planar position was adapted.

The invariant pulses (vertical and orientation). Because $\delta_z = 0$, the TCP height never changes, so $\dot{z}$ (blue) shows the same three pulses every cycle — -0.41 m/s at the grasp, -1.50 m/s at the assembly descent, $+1.64$ m/s at the retraction — and the mirror $\dot{q}_3$ (yellow) is likewise unchanged. Equally, because the orientation is *not* adapted in Requirement 2 ($\theta_\delta = 0$), the 90° screw is the same every cycle, so the deep wrist dip $\dot{q}_4 = -\dot{\varphi} \approx -3.10$ rad/s at the assembly recurs identically. These fixed pulses confirm that the z and φ degrees of freedom were left untouched.

Transfer spike changes. The large joint-rate spikes at the middle of the transfer ($\dot{q}_1$ and $\dot{q}_4$, the Jacobian amplification near the folded posture) scale with the goal: their peaks are ≈ 3.1 rad/s for $\boldsymbol{\delta}_1$, rise to ≈ 4.2 rad/s for $\boldsymbol{\delta}_2$, and fall to ≈ 1.8 rad/s for $\boldsymbol{\delta}_3$. The mechanism is twofold. First, the transfer Cartesian velocity is the demonstrated profile multiplied by the spatial scale $\mathbf{S}_g$, so a farther goal ($\boldsymbol{\delta}_2$, $S_x = 1.32$) means a faster sweep and a nearer goal ($\boldsymbol{\delta}_3$, $S_x = 0.36$) a gentler one — visible already in $\dot{y}$, whose transfer peak goes $1.10 \to 1.21 \to 1.02$ m/s. Second, this Cartesian velocity is amplified by the Jacobian, and the *amount* of amplification depends on where the shifted path travels: $\boldsymbol{\delta}_2$ reaches into a more folded, worse-conditioned region (its path comes closest to the base, min $\sqrt{x^2+y^2} \approx 0.31$, $q_2 \approx 162°$), while $\boldsymbol{\delta}_3$ stays in a better-conditioned one. The two effects compound, so the *joint* spikes vary more ($\approx \pm 35\%$) than the Cartesian speeds ($\approx \pm 10\%$) — the nonlinear Jacobian effect on top of the linear scaling. *Note:* for $\boldsymbol{\delta}_2$ the peak (≈ 4.2 rad/s) exceeds the ≈ 3 rad/s nominal joint-rate guideline; this occurs only transiently near the folded posture, and $\boldsymbol{\delta}_2$ ($\delta_x = -0.2$) is itself outside the stated range $\delta_x \in [-0.1, 0.5]$ of the statement.

Return spike changes. The tall $\dot{x}$ pulse of the fly-by (+1.57, +1.91, +0.89 m/s for the three cycles) also tracks $\boldsymbol{\delta}$, but for a different reason: the retract point P_{RETRACT} sits above the shifted assembly station, whereas the via point $P_{\text{VIA_RET}}$ and home P_0 are fixed. The outward leg therefore spans $\Delta x = 1.0 - g_{P_4,x} = 0.92, 1.12, 0.52\,\text{m}$ in the three cycles, so its peak velocity grows for $\boldsymbol{\delta}_2$ (the part is delivered farthest in $-x$, so the return is longest) and shrinks for $\boldsymbol{\delta}_3$. The corresponding $\dot{y}$ return excursion shifts in the same way.

Summary Every velocity still begins and ends at zero in each cycle and joins continuously across the $t = 12, 24\,\text{s}$ boundaries, so the motion stays smooth and periodic. What changes from cycle to cycle is exactly the set of pulses belonging to the adapted x, y coordinates — the transfer amplification and the return fly-by — and the size of each change is a direct, predictable readout of the corresponding $\boldsymbol{\delta}$ through the spatial scale $\mathbf{S}_g$, while the vertical and orientation pulses stay fixed.

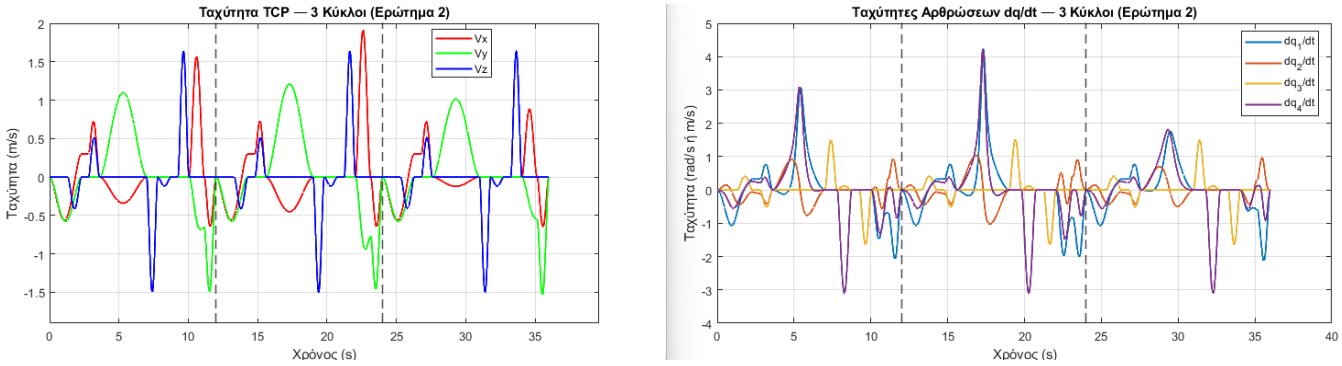


Figure 3: Requirement 2 — TCP Cartesian velocity $(\dot{x}, \dot{y}, \dot{z})$ and joint rates $\dot{q}_{1\text{–}4}$ over three cycles. The $\dot{z}, \dot{q}_3$ and the assembly $\dot{q}_4$ pulses are identical every cycle (vertical/orientation untouched); only the transfer $\dot{q}_1, \dot{q}_4$ spikes and the return fly-by $\dot{x}$ pulse vary with $\boldsymbol{\delta}$.

Requirement 3 – Orientation Adaptation with DMP

Requirement 3 repeats the three position scenarios of Requirement 2 ($\boldsymbol{\delta}_1 = \mathbf{0}$, $\boldsymbol{\delta}_2 = [-0.2, 0.2, 0]$, $\boldsymbol{\delta}_3 = [0.4, -0.15, 0]$) but now *also* adapts the orientation with a DMP, using the initial conditions $\theta_{\delta_1} = 0$, $\theta_{\delta_2} = 20°$, $\theta_{\delta_3} = -25°$. Since the camera detects part B rotated about z by θ_δ, part A must be screwed at the matching angle

$$\varphi_{\text{goal}} = \tfrac{\pi}{2} + \theta_\delta, \tag{24}$$

so the orientation goals of the three cycles are 90°, 110° and 65°.

Orientation as a scalar

By Hint 1 and the kinematics of Part 1, the manipulator can rotate the tool only about the vertical $-z$ axis, and the whole orientation is captured by the single angle $\varphi = q_1 + q_2 - q_4$ (Part 1, Eq. 4). The orientation therefore lives in a one-dimensional space, and adapting it needs nothing more than an ordinary *scalar* DMP acting on φ: the general quaternion-based orientation DMP collapses to the scalar position-DMP formulation of Requirement 1 applied to the single coordinate φ. No quaternion logarithms or 3×3 scaling matrices are required.

Training the orientation DMP

The demonstration is the Part 1 screw motion (Segment 5b): the quintic $\varphi_d : 0 \to \frac{\pi}{2}$ executed in $\tau_B = T_d = 0.95\,\mathrm{s}$, together with its derivatives $\dot{\varphi}_d, \ddot{\varphi}_d$. Specialising the training of Requirement 1 to the one coordinate φ, the demonstrated forcing is

$$f_d(t) = \tau_B^2\,\ddot{\varphi}_d(t) - \alpha_z\big(\beta_z(\varphi_{gd} - \varphi_d(t)) - \tau_B\,\dot{\varphi}_d(t)\big), \qquad \varphi_{gd} = \tfrac{\pi}{2}, \tag{25}$$

and the weights w_i are obtained by the same Locally Weighted Regression (Eq. 9 of Requirement 1), with the canonical system $x(t) = \exp(-(\alpha_x/\tau_B)t)$ and the Gaussian centres/widths placed by Eq. 7. This orientation DMP is independent of the position DMP: it has its own canonical system and runs over the 0.95 s assembly screw window $[7.8, 8.75]$, whereas the position DMP runs over the 3.4 s transfer $[3.6, 7.0]$; the two share only the gain values.

Spatial scaling to the detected orientation

At execution the start is $\varphi_0 = 0$ and the goal is $\varphi_{\mathrm{goal}} = \frac{\pi}{2} + \theta_\delta$, so the (scalar) spatial scale is

$$S_\varphi = \frac{\varphi_{\mathrm{goal}} - \varphi_0}{\varphi_{gd} - \varphi_{0d}} = \frac{\frac{\pi}{2} + \theta_\delta}{\frac{\pi}{2}} = 1 + \frac{2\,\theta_\delta}{\pi}. \tag{26}$$

The three cycles give $S_\varphi = 1.000,\ 1.222,\ 0.722$. Because $\varphi_{\mathrm{goal}} \in [65^\circ, 110^\circ]$ is strictly positive and hence $S_\varphi > 0$ throughout the admissible range $\theta_\delta \in [-30^\circ, 30^\circ]$, there is no trajectory flipping: the screw is always a smooth, monotone forward rotation, only ending at a different final angle. The DMP stretches the demonstrated profile for $\theta_\delta > 0$ (cycle 2) and compresses it for $\theta_\delta < 0$ (cycle 3).

Coupling with the position through the wrist joint

The wrist angle $q_4 = q_1 + q_2 - \varphi$ now carries *both* adaptations: the arm angle $q_1 + q_2$ follows the position offset $\boldsymbol{\delta}$ (through the inverse kinematics at the shifted station) and φ follows the orientation offset θ_δ. During the assembly the planar coordinates are held, so $q_1 + q_2$ is constant and the wrist alone produces the screw,

$$\Delta q_4 = -\Delta\varphi = -\Big(\tfrac{\pi}{2} + \theta_\delta\Big), \tag{27}$$

i.e. a wrist rotation of -90°, -110°, -65° in the three cycles — no longer the fixed -90° of Requirement 2. Table 3 collects the resulting quantities.

Table 3: Orientation adaptation per cycle (g_{P_4} includes $\boldsymbol{\delta}$).

Cycle	θ_δ	φ_{goal}	S_φ	$q_1{+}q_2$ at g_{P_4}	$q_4(P_4)$	screw Δq_4
1	0°	90°	1.000	145.3°	55.3°	-90°
2	$+20^\circ$	110°	1.222	148.6°	38.6°	-110°
3	-25°	65°	0.722	121.3°	56.3°	-65°

Changes relative to Requirement 2

In Requirement 2 the orientation was not adapted, so the screw was identical in every cycle and the wrist-rate dip at the assembly was a fixed $\dot{q}_4 = -\dot{\varphi} \approx -3.10\,\mathrm{rad/s}$. In Requirement 3 the

orientation DMP scales the screw in time as well: with $\tau = \tau_B$ unchanged, the peak angular rate is

$$\dot{\varphi}_{\max} = S_\varphi \, \dot{\varphi}^{\text{demo}}_{\max} = S_\varphi \cdot \tfrac{15}{8} \, \frac{\pi/2}{\tau_B} \approx S_\varphi \cdot 3.10 \text{ rad/s}, \tag{28}$$

so the assembly wrist dip now *varies* per cycle: $\dot{q}_4 \approx -3.10, -3.78, -2.23\,\text{rad/s}$ (deeper for $\theta_\delta > 0$, shallower for $\theta_\delta < 0$). This is the clearest new signature: the assembly-window wrist pulse, which was invariant in Requirement 2, becomes cycle-dependent because the orientation is now adapted. Correspondingly the return begins from the shifted assembly orientation φ_{goal} and is brought back to the fixed home value through the Part 1 return quintic.

Parameters and integration

The orientation DMP uses the same gains as the position DMP — $\alpha_z = 25$, $\beta_z = \alpha_z/4$, $\alpha_x = 1.0$, $N = 40$ Gaussians — with temporal scale $\tau = \tau_B = 0.95\,\text{s}$ (the assembly screw duration) and Euler step $\Delta t = 10^{-3}\,\text{s}$ (Hint 2). Keeping $\tau = \tau_B$ preserves the 0.95 s screw duration of Part 1, so the overall 12 s cycle timing is unchanged.

Desired and Produced Orientation

The required common plot superposes the demonstration $\varphi_d : 0 \to 90°$ (dashed) and the three roll-outs ending at $90°, 110°, 65°$. As for the position DMP, all curves share the same start $\varphi_0 = 0$ with zero initial slope, keep the demonstrated smooth S-shape, and converge with zero terminal slope to their goals; cycle 1 ($S_\varphi = 1$) lies exactly on the demonstration, validating the training, while cycles 2 and 3 are the predictable stretched and compressed versions.

Position/Joint Trajectories with Adapted Orientation

The three cycles use the same position offsets as Requirement 2 plus the orientation offsets $\theta_\delta = 0$, $20°$, $-25°$, giving screw goals $\varphi_{\text{goal}} = 90°, 110°, 65°$.

The Cartesian position is unchanged from Requirement 2. Adapting the orientation does not move the TCP: the angle φ is a separate coordinate, so the x, y, z plot here is *identical* to that of Requirement 2. The height z (blue) repeats the same grasp dip to 1.63 and assembly dip to 1.15 every cycle ($\delta_z = 0$), while x (red) settles at $0.08, -0.12, 0.48\,\text{m}$ and y (green) at $1.0, 1.2, 0.85\,\text{m}$ — exactly as before. Consequently the joints driven by position alone are also unchanged: q_1 (blue), q_2 (orange) follow the same plateaus as in Requirement 2 and q_3 (yellow), set by z, is bit-for-bit identical.

The only new behaviour is in the wrist q_4. Since $q_4 = q_1 + q_2 - \varphi$ and the planar part $q_1 + q_2$ is held during the assembly, the wrist alone carries the orientation adaptation. Its assembly-hold value becomes

$$q_4(P_4) = \underbrace{(q_1 + q_2)}_{\text{from } \boldsymbol{\delta}} - \underbrace{\varphi_{\text{goal}}}_{\frac{\pi}{2} + \theta_\delta},$$

i.e. 55.3°, 38.6°, 56.3° ($\approx 0.97, 0.67, 0.98\,\text{rad}$) in the three cycles, and the wrist "step" performed during the screw is now $\Delta q_4 = -(90° + \theta_\delta) = -90°, -110°, -65°$, no longer a fixed $-90°$. The transfer part of q_4 (where $\varphi = 0$, so $q_4 = q_1 + q_2$) is unaffected and keeps its Requirement 2 peaks $\approx 2.5, 2.6, 2.1\,\text{rad}$.

Brief comparison with Requirement 2. The two questions differ in exactly one signal. In Requirement 2 the wrist held $q_4(P_4) = 55.3°, 58.6°, 31.3°$ (always a $-90°$ screw); in Requirement 3 the same plateaus shift to $55.3°$, $38.6°$, $56.3°$. The shift is precisely

$$q_4^{(3)}(P_4) - q_4^{(2)}(P_4) = -\theta_\delta = 0,\ -20°,\ +25°,$$

so a part rotated positively ($\theta_{\delta_2} = +20°$) lowers the wrist by $20°$, while a negatively rotated one ($\theta_{\delta_3} = -25°$) raises it by $25°$. This makes the decoupling explicit: the position offset $\boldsymbol{\delta}$ lives entirely in q_1, q_2, q_3 (and hence in the TCP plot), whereas the orientation offset θ_δ lives entirely in q_4. Everything else — the collision-free shape, the return through the fixed via point, and the clean periodic return to P_0 at $t = 12, 24\,\text{s}$ — is preserved, so Requirement 3 adds the orientation adaptation on top of Requirement 2 without disturbing any of the position behaviour.

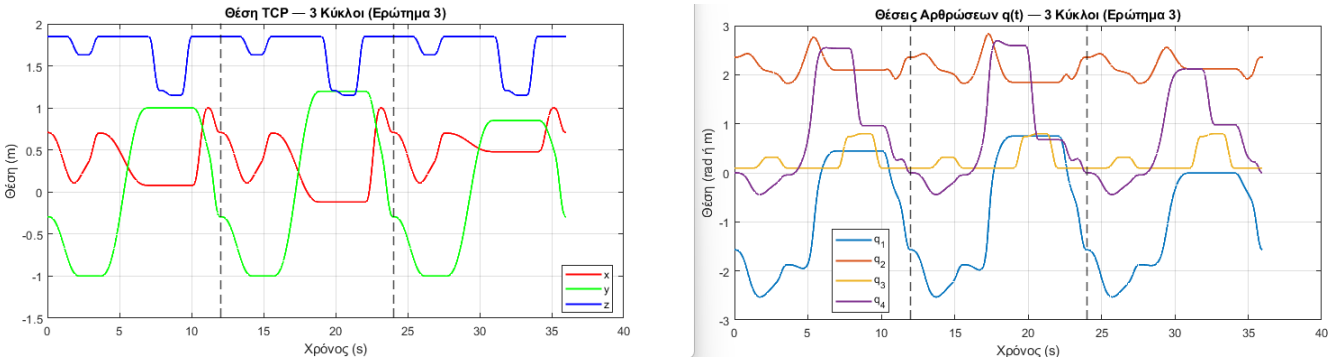


Figure 4: Requirement 3 — TCP position and joint coordinates over three cycles with both position *and* orientation adapted ($\theta_\delta = 0, 20°, -25°$). The x, y, z traces and q_1, q_2, q_3 are bit-for-bit those of Requirement 2; only the wrist q_4 assembly-hold plateau changes, by exactly $-\theta_\delta$ ($0, -20°, +25°$ relative to Requirement 2).

Reading the orientation DMP figure

This is the orientation counterpart of the position desired-vs-produced figure: over the $0.95\,\text{s}$ assembly screw window it superposes the training demonstration $\varphi_d : 0 \to \pi/2$ (dashed) and the three roll-outs for $\theta_\delta = 0, 20°, -25°$, whose goals are $\varphi_{\text{goal}} = \frac{\pi}{2} + \theta_\delta = 90°, 110°, 65°$ ($= 1.571,\ 1.920,\ 1.134\,\text{rad}$).

Reproduction (validation). For $\theta_{\delta_1} = 0$ the scale is $S_\varphi = 1$, so the red curve lies exactly on the dashed desired and ends at $\pi/2$. This confirms that the scalar orientation DMP is trained correctly — it regenerates the Part 1 screw before any adaptation.

Reading the $20°$ (and $-25°$) change straight off the curve. Because $\varphi_{\text{goal}} = \frac{\pi}{2} + \theta_\delta$, the *final value* of each φ curve *is* the screw angle, and its vertical gap from the dashed demonstration's $90°$ endpoint is exactly θ_δ. The blue curve terminates at $1.920\,\text{rad} = 110°$, i.e. $20°$ above the dashed $90°$ endpoint: that gap read off the φ-axis at $t = \tau_B$ *is* $\theta_{\delta_2} = +20°$. The green curve ends at $1.134\,\text{rad} = 65°$, $25°$ *below* the dashed endpoint, i.e. $\theta_{\delta_3} = -25°$. One need not even wait for the endpoint: since each roll-out is a *constant multiple* of the demonstration, the ratio of any two simultaneous ordinates already gives the scale. At the screw midpoint ($t \approx 0.475\,\text{s}$) the dashed curve passes through $45°$ ($\pi/4$), the blue through $55°$ ($0.96\,\text{rad}$) and the green through $32.5°$ ($0.57\,\text{rad}$); hence $S_\varphi = 55/45 = 1.222$ and $32.5/45 = 0.722$, and $\theta_\delta = (S_\varphi - 1) \cdot 90° = +20°, -25°$ — recovered from the *interior* of the curve, not just its end. The visual fingerprint of this pure proportional scaling is that the three curves *fan out linearly* from a common origin: zero gap at $t = 0$, the gap widening monotonically to its maximum at $t = \tau_B$. Had the orientation *not*

been adapted (Requirement 2), all three roll-outs would collapse onto the dashed demonstration and the gap would be zero everywhere — so the very existence of a non-zero, growing gap is the graphical proof that the orientation was changed, and its terminal height quantifies by how much. Finally, the flat (zero-slope) start and finish of every curve show the screw still begins and ends at rest, so it splices smoothly onto the rim descent before it and the release hold after it.

Exact proportional scaling. Because the screw starts at $\varphi_0 = 0$ (and $\varphi_{0d} = 0$), the spatial scale reduces to $S_\varphi = \varphi_{\text{goal}}/(\pi/2)$ and the roll-out is an *exact scalar multiple* of the demonstration at every instant,

$$\varphi_{\text{DMP}}(t) = S_\varphi \, \varphi_d(t).$$

This follows directly from the DMP equations: substituting $\varphi = S_\varphi \varphi_d$ (hence $z = S_\varphi z_d$) reproduces the demonstrated dynamics with the forcing multiplied by S_φ, which is precisely $\mathbf{S}_g = S_\varphi$ when the start is the origin. This is cleaner than the position case, where a non-zero start makes the scaling affine rather than a pure multiple — which is exactly why the position curves of Requirement 2 share a common *start* and fan out only towards the goal, whereas here the curves are proportional over their *whole* length.

Shape, rate and convergence. All curves inherit the demonstration's smooth quintic S-shape with *zero slope at both ends*, so each screw begins and finishes at rest — continuous with the descent that precedes it and the release hold that follows. Differentiating the proportionality, the angular *rate* also scales by S_φ, so the peak $\dot{\varphi}$ is $3.10 \times \{1, 1.222, 0.722\} = \{3.10, 3.78, 2.23\}$ rad/s; this is exactly what makes the assembly wrist-rate dip $\dot{q}_4$ cycle-dependent in the velocity plots, in contrast to its fixed value in Requirement 2. Each roll-out converges with zero terminal slope to its goal at $t = 0.95\,\text{s}$ $(90°, 110°, 65°)$, so the part is screwed to precisely the camera-detected angle every time, monotonically and without overshoot. Since $\varphi_{\text{goal}} \in [65°, 110°] > 0$ (hence $S_\varphi > 0$) over the whole admissible range $\theta_\delta \in [-30°, 30°]$, no curve ever flips or reverses.

Summary. The figure exhibits the two defining DMP properties for the orientation: exact reproduction of the demonstration at $\theta_\delta = 0$, and predictable, purely proportional generalisation for $\theta_\delta \neq 0$, delivering the correct screw angle robustly under the $\pm 30°$ orientation uncertainty with no loss of smoothness.

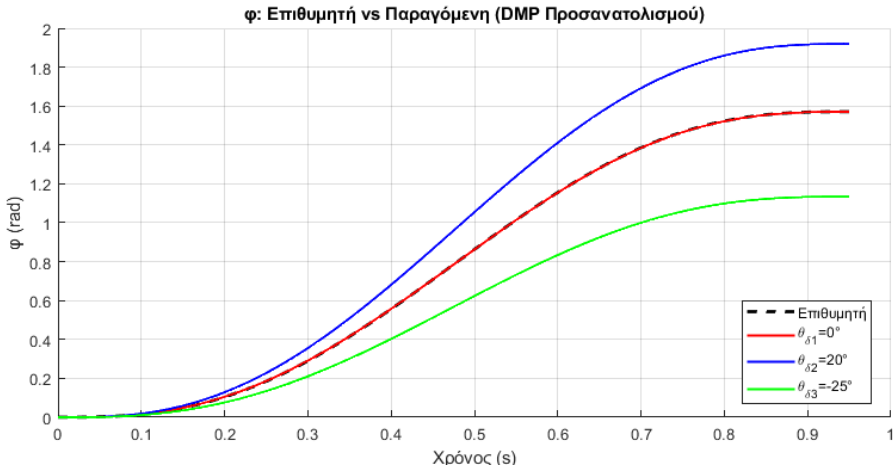
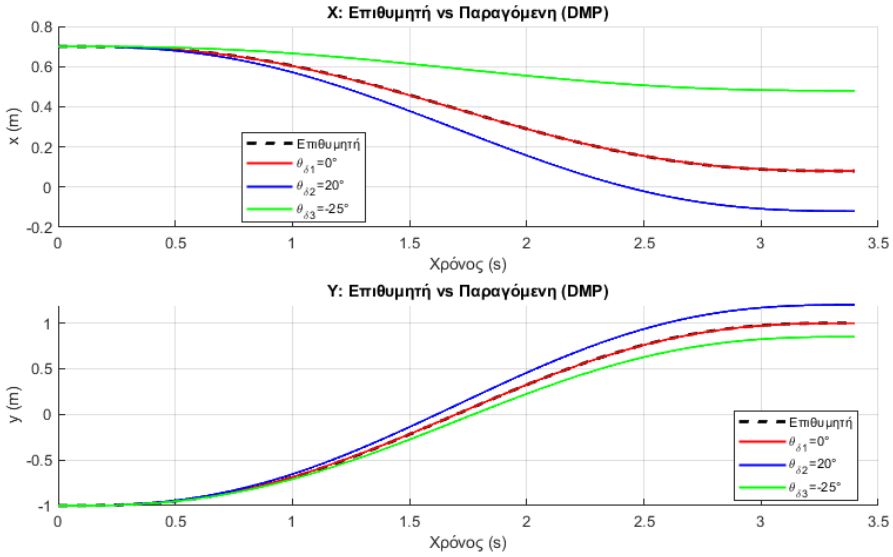


Figure 5: Requirement 3 — Orientation DMP, desired (dashed, $\varphi_d : 0 \to 90°$) vs. generated for $\theta_\delta = 0, 20°, -25°$ over the 0.95 s screw. The roll-outs end at $90°, 110°, 65°$: the terminal gap from the dashed 90° is exactly θ_δ, and each curve is a constant $S_\varphi = 1.000/1.222/0.722$ multiple of the demonstration.



TCP orientation φ over the three cycles

The figure plots the tool angle φ (the fourth task coordinate) across the three cycles, with the dotted lines marking the three screw goals $\varphi_{\text{goal}} = \frac{\pi}{2} + \theta_\delta = 90°, 110°, 65°$ ($1.571, 1.920, 1.134$ rad).

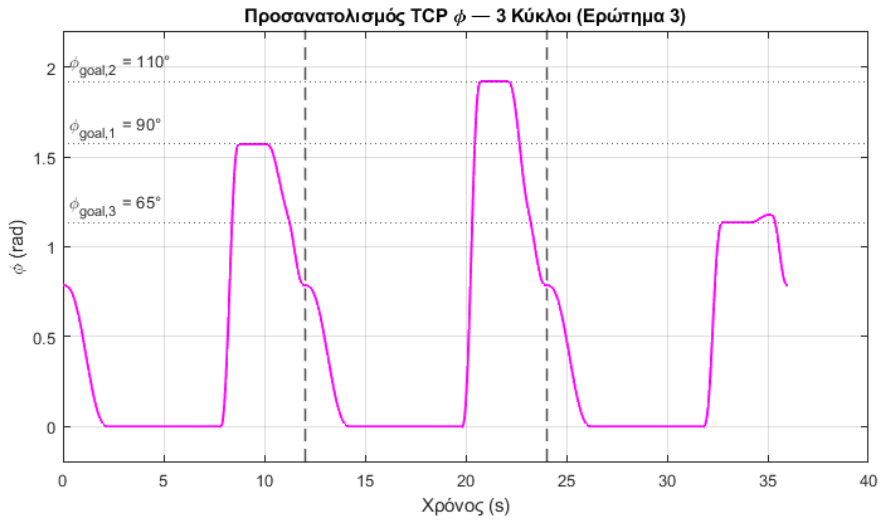
The three screws lock onto their detected goals. In each cycle φ rises from 0 to $\frac{\pi}{2} + \theta_\delta$ during the $[7.8, 8.75]$ screw window, and the plateau sits exactly on the corresponding dotted line: 90° in cycle 1, 110° in cycle 2 and 65° in cycle 3. The height of each plateau above the 90° reference is therefore the injected rotation θ_δ, so the $+20°$ and $-25°$ camera offsets can be read directly off the plot.

Held orientation during transport, screw confined to the assembly. Between $t \approx 2.3$ and 7.8 s of every cycle φ is flat at 0: the part keeps its grabbed orientation through grasp, lift, transfer and the rim descent, and the 90°-class rotation happens only in the 0.95 s assembly window. This confirms that the orientation DMP acts solely on the screw segment and leaves the rest of the motion untouched.

Clean periodic return to the home orientation. Each cycle starts and ends at the home value $\varphi = \pi/4 \approx 0.785$ rad: at $t = 0, 12, 24$ s the curve passes through $\pi/4$ and then relaxes to 0 during the approach. The brief flat point at $\pi/4$ on the cycle boundaries is the rest-to-rest condition ($\dot{\varphi} = 0$ at P_0), which is what makes the three cycles join seamlessly and the process cleanly periodic despite the different screw angles.

Return through the fixed via orientation $3\pi/8$. On the way home φ is brought from the screw goal to P_0 through the fixed return via value $3\pi/8 = 67.5°$. Because this via angle lies above the cycle-3 goal (65°) but below the cycle-1/2 goals (90°/110°), the return is monotonically *downward* in cycles 1 and 2, whereas in cycle 3 φ first rises slightly from 65° to 67.5° (the small upward step visible near $t \approx 34$ s) before descending to 45°. This matches the per-cycle via displacements $-22.5°, -42.5°, +2.5°$.

Continuity. Every transition has continuous slope — no vertical jumps — so the orientation trajectory is C^1 throughout, in line with the continuous-velocity requirement.



Velocity Profiles across the three cycles

The TCP velocity is identical to Requirement 2. Since the orientation does not enter the Cartesian position, the x, y, z motion — and therefore $\dot{x}, \dot{y}, \dot{z}$ — is exactly that of Requirement 2. The plot reproduces the same features cycle by cycle: $\dot{z}$ (blue) with the invariant grasp/assembly/retract pulses ($-0.41, -1.50, +1.64$ m/s, since $\delta_z = 0$); the $\dot{x}$ (red) transfer lobe and the return fly-by spike that grows with the delivery distance ($+1.57, +1.91, +0.89$ m/s); and the $\dot{y}$ (green) transfer bell scaled by S_y ($1.10, 1.21, 1.02$ m/s). Nothing in the end-effector velocity changes relative to Requirement 2.

Joint rates: only $\dot{q}_4$ changes, and only at the assembly. The rates driven by position alone are unchanged from Requirement 2: $\dot{q}_1, \dot{q}_2$ (from x, y) and $\dot{q}_3 = -\dot{z}$ (yellow, the $+1.50/-1.64$ pulses) are identical. Even the large *transfer* spike of $\dot{q}_4$ is unchanged, because during the transfer $\varphi = 0$ and hence $\dot{q}_4 = \dot{q}_1 + \dot{q}_2$ — its Jacobian-amplified peaks are the same $\approx 3.1, 4.2, 1.8$ rad/s as before. The single new effect appears in the *assembly* dip of $\dot{q}_4$.

The assembly wrist dip now scales with θ_δ. During the assembly the planar coordinates are held, so $\dot{q}_1 = \dot{q}_2 = 0$ and $\dot{q}_4 = -\dot{\varphi}$. With the orientation DMP scaling the screw by S_φ, the peak rate is $\dot{\varphi}_{\max} = S_\varphi \cdot 3.10$ rad/s, giving

$$\dot{q}_4^{\text{assembly}} \approx -3.10,\ -3.78,\ -2.23 \text{ rad/s} \quad \text{for } \theta_\delta = 0,\ 20°, -25°,$$

i.e. a deeper dip for the larger screw (110°, cycle 2) and a shallower one for the smaller screw (65°, cycle 3). On the plot this is read off directly: the depth of the assembly $\dot{q}_4$ dip relative to the $\theta_\delta = 0$ reference is in the same $1 : 1.222 : 0.722$ ratio as the screw angles, so the dip is the velocity-domain echo of the $+20°/-25°$ change seen as a level shift in the position plot. The return $\dot{q}_4$ also shifts slightly between cycles, because the homeward φ leg now starts from the different goal φ_{goal} (its displacement to the fixed $P_{\text{VIA_RET}}$ at 67.5° is $-22.5°, -42.5°, +2.5°$, the last almost zero).

Comparison with Requirement 2. The two questions differ in exactly one place. In Requirement 2 the assembly dip of $\dot{q}_4$ was a fixed -3.10 rad/s in every cycle (the screw was always 90°); in Requirement 3 it varies as $-3.10/-3.78/-2.23$ rad/s. Everything else — the whole TCP velocity, $\dot{q}_1, \dot{q}_2, \dot{q}_3$, and the transfer spike of $\dot{q}_4$ — is bit-for-bit the same. This cleanly isolates the orientation adaptation: it adds a cycle-dependent rate only on the wrist and only during the 0.95 s screw, leaving the position dynamics untouched, and every signal still returns to zero at the cycle boundaries $t = 12, 24$ s for a smooth periodic execution.

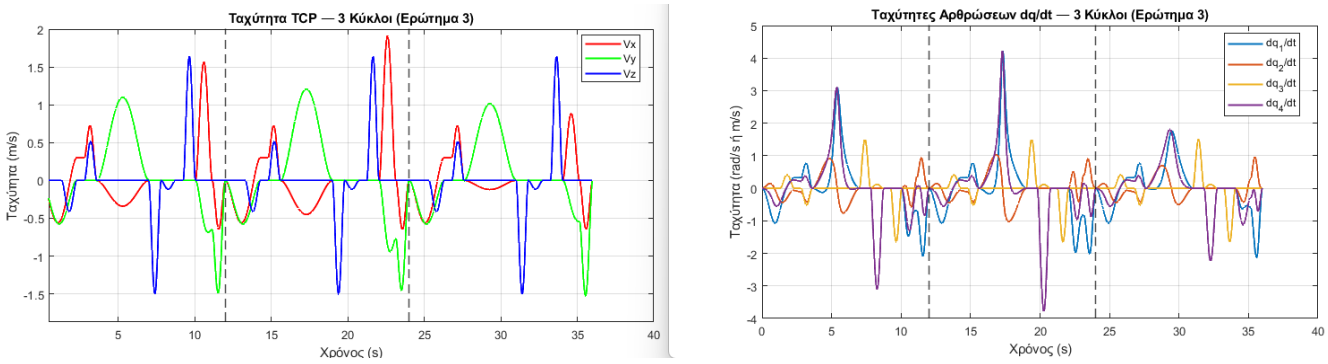


Figure 6: Requirement 3 — TCP and joint velocities over three cycles. Every trace matches Requirement 2 except the assembly dip of $\dot{q}_4$, which now scales with S_φ ($-3.10/-3.78/-2.23\,\mathrm{rad/s}$) — the velocity-domain signature of the adapted orientation.